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CODE BOOK FOR CASHORE'S 4PT FRAMEWORK

Benjamin Cashore[1] with contributions from many [2]

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A. THE CODING TASK

This codebook develops a systematic and standardized way to classify qualitative documents into one of Cashore's four problem types(4PT) framework. It will do so by providing guidance to coders on how to assign Type 1, 2, 3 or 4 status to qualitative text including, but not limited to, peer reviewed journal articles, course syllabi, policy statements, policy analyses, media communications and policy recommendations. This classification effort complements, rather than replaces, careful qualitative application of the framework required for uncovering important nuances through deliberative analysis and reflection of highly complex text. ***What the quantitative coding effort does well is to generate useful***

knowledge about whether the overall patterns identified by an one study or set of studies compare to other cases outside of the study. Coding also permits assessments as to whether there are differences in the dominance of one or more problem types across time and also space – be they countries, organizations, or sectors.

Hence, not only can a well-developed code book generate descriptions about a range of policy and governance outcomes, it will also allow for broadening epistemological and methodological techniques that might explain variation – such as the correlation between the kind of disciplinary, sub-disciplinary and interdisciplinary training experts received that shape their bias in emphasizing one Type over the other. This in turn, might help generate knowledge about, and reverse when desired, "unconscious bias" about the way policy scholars and practitioners conceive of, and measure, "on the ground" problems. In other words, this framework can help advance "fit for purpose" policy analysis.

To be clear: unlike Cashore's broader argument about the drift away from Type 4 problems that he has, and is, applying to understand their role in shaping sustainability policy and governance over the last 35 years, the code book is designed to describe a range of outcomes, to which Cashore and his collaborators, and anyone else for that matter, can better assess, test, and develop their own arguments and theories. In short, the code book, paves the way for, but does not take a position on, which of the four problem types is preferred nor the causes of changes in emphasis of the Types.

Importantly, although the 4PT distinctions originally came out of Cashore's work on global, transnational and local approaches to sustainability policy and governance, they can be applied to disentangle how any kind of problem facing society and policy making are analysed, and treated, by scholars and experts. This means that ***while Cashore applies and draws examples from sustainability policy scholarship and practice, the framework is equally useful for anyone working on problem focused policy analysis including, but not limited to those scholars working in schools of public health, education, social work, forestry, fisheries, natural resources, agriculture, and even mining and energy for that matter.*** Having said this, the guidance and examples below for coders to be able to render concrete important but sometimes abstract distinctions, draw on examples from five years of efforts to code sustainability policy and governance analyses.

This code book proceeds in the following steps. First, we discuss Cashore's "utility" and "problem solving" labels as the two key distinctions that position the 4PT framework. Second, we review Cashore's four key schools of sustainability thought and practice – including notable strands within them – to show how each of these 4PTs are subtly nurtured by sophisticated scholarly communities. Third, we review the challenges facing coding of different kinds of qualitative documents. Fourth, we review different sorting challenges based on the kinds of document being including abstracts of course syllabi, journal articles. Fifth we offer solutions to each including reviewing of our protocol for sorting scholarly articles through careful qualitative analysis that has been developed and revised over the last five years. We then review our approach to coding course syllabi and the development of key words to facilitate swift coding of a range of documents. We then discuss our systematic efforts to advance AI assisted coding with the goal of producing systematic text analysis that would be consistent with the laborious human coding efforts of the last six years (i.e. we seek to achieve intercoder reliability between AI and humans .8 or better). We conclude by offering recommendations on next steps for both Cashore's team and collaborators, as well as those who might want to apply the 4PT framework beyond sustainability policy and governance considerations. Appendices A reviews our approach to coding articles beyond the two distinctions as a way to generate propositions about preliminary explanations for results and Appendix B which provides examples from cases with which to assist in sorting initially "ambiguous" cases.

a) Key distinctions: utility and problem solving orientations

The specific coding task appears at first glance seems quite straightforward i.e. to distinguish whether scholarly and practitioner policy analyses emphasise, or not, "utility" and "problem solving" orientations (Table 1).

Table 1: The Four Problem Types (4PT)- Simplified Distinctions			
		Utility dominates?	
		YES	**NO**
Problem solving dominates?	**YES**	**TYPE 1**	**TYPE 4**
	NO	**TYPE 2**	**TYPE 3**

These two distinctions originally emerged from Cashore's reflections, individually and with colleagues, about why so many sustainability policy experts have disagreed for over 35 years about how to think about, and also manage, critical environmental challenges facing the planet highlighted by climate, biodiversity and forest degradation. (Again, while derived from the sustainability policy and governance cases, the 4PT distinctions are meant to be applied to any "on the ground" problems facing policy makers and society including health care, education, child care, education and so on).

The origins of the 4PT distinctions followed Cashore's recognition, having only worked in interdisciplinary forestry, environment and policy schools, that longstanding largely unconscious biases among himself, his colleagues, and experts generally, about how to study the world of sustainability policy and governance, helped explain longstanding debates about not only about how to conceive of, and measure, sustainability challenges, but also recommended courses of policy action for managing the environment, including climate, species extinctions, biodiversity loss, as well as forests and natural resources management.

What Cashore noticed was two key divides within these debates. ***The first distinction*** was whether sustainability scholars and practitioners believed that enhancing **"utility"** outcomes was the key task facing sustainability policy analysis. He noted that a large group of scholars generally felt that the more utility was enhanced, the better off society will be. While this causal belief is widely diffused among a range of analysts, it is also hotly debated by scholars and analysts who either oppose utility as a guiding orientation, and/or seek other ways to frame desired outcomes. Hence, Cashore recognized that the belief that utility enhancement was the preferable, from those that did not, was an excellent way to distinguish sustainability policy and governance analyses.

Cashore's second distinction came from his recognition that while some narrowed their conclusion to assessing how to solve or effectively manage the clearly specified problem at hand- such as such as saving a particular species from extinction, avoiding resource collapse of a specified fish stock, or limiting warming to 1.5 degrees above pre-industrial levels – others brought in analytical frames that allowed for a deliberations over not only how, but ***whether*** the problem at hand could or should be solved. That is, this latter group applied adjudicating frameworks that were developed as a way to address ***any*** particular problem – regardless as to its unique features - that might find its way to the societal or policy agenda in mind. Rather than generating research and analysis to shed light on how to manage a specific policy challenge, this group was united by developing frameworks that allow for consideration of not acting at all, or acting in a way that does not effectively manage the problem at hand.

b) Conceptual foundations: utility and problem contingent labels

While these distinctions showed great promise, the next task was to elaborate their meanings so that their influence could be better identified and assessed.

The utility label covers a broad range

Cashore's 4PT framework notes that the term "utility" is generally applied and defined broadly. It includes for example, those trained in welfare economics who, largely focusing on income or economic growth, convert others outcomes to "utility equivalents" in order to monetize, and compare, human preferences for green space and nature, with their direct interest in generating income. The term "utility" also incorporates the broader philosophy of "utilitarianism" that seeks to advance outcomes that produce the "greatest good for the greatest number in the long run". It is also used more generally as a term to discuss whether, in terms of enhancing individual or collection satisfactions, there is any "utility" in a proposed course of action?

For Cashore, the rationale for use of "utility" as a label was that it best captures the general belief behind a range of analyses that their goal is to improve "**human satisfactions".** This is an important clarification because the utility as a term has a decidedly anthropogenetic orientation in which policy is geared towards improving the interests of humans either today or in the future. This, in turn, is shaped by some kind of homage and widespread work developing theories relevant to unpacking human **"self-interest"**. What is important for coding on the 4PT framework is that the concern itself is what is important – i.e. **the utility label in the 4PT framework is shorthand to capture those who are motivated to advance human self-interested satisfactions that incorporate a range of more specific satisfaction seeking rationales**

This orientation in turn, allows coders to identify efforts to promote and understand **self-interested behaviours at multiple levels** of sustainability policy and governance including individual, organizational, nation state decisions as well as those within international governance arenas and global governance agreements. Hence, the utility label includes, for coding purposes, analyses focused on aggregating **individual utility**, **organizational utility** (such as companies returning shareholder value), or even **states' utility** by maximizing their own self-interested preferences. This category incorporates, rather than distinguishes, longstanding debates among neo-realist and neo-liberal institutionalist scholars within international relations about whether self-interested, utility chasing states, are motivated to achieve relative or absolute gains. (The 4PT framework notes that debates about relative or absolute gains have undermine the similar quest to advance utility seeking interests, including, at the state level, relying on some conception of "national interest" that improves the satisfaction of its population through some kind of beneficial "rational" outcome.

Hence, this broad use of the utility label focuses coders on identifying any kind of analysis that is motivated by enhancing human satisfaction through self-interested preference maximization. This allows coders to capture in one category a range of scholarship and analysis that although diverse and nuanced, are united by their underlying orientation in emphasizing individual, organizational and state based preferences that pay homage to improving human "satisfactions" at multiple levels.

The problem solving label: distinguishing analyses that target "how", not "whether"

The term "problem solving" is also used broadly to capture a range of expertise and approaches that undertake research with which to solve or effectively manage a particular "on the ground" problem. That is, they eschew asking "whether" a problem can be effectively managed, and instead ask, given the problem at hand, how can it be effectively managed?

This category then incorporates a broad group of scholars and analysts who are united in their quest to understand and effectively management an "on the ground" problem such as the extinction of a particular species or limiting warming to no more than 1.5/2 °C above pre-industrial levels promoted through the Paris Accord. This means that the analyses they bring, and conclusions they make, will go back to implications for generating knowledge useful for problem solving of a problem, or a class of problems. Similarly, these analysts will never apply universal pre-established theories to assess whether the problem at hand ***ought*** to be solved. This is not their interest. Likewise while their research may carry lessons for other problems, they do not seek to generate universal theories that can be applied to any kind of problem.

Put another way, this group of scholars would, for these reasons, avoid any quest for developing a theoretical orientation, or set of recommendations, that went beyond the problem or class of problems from which their research was derived. The practical result is that many in this category tend to carefully assess problem structure or "key features of the problem" they are attempting to shed light on how to better manage or solve.

This is important. The reason they eschew any effort to develop a universal way to address any problem is because, by disconnecting from problem structure, the approach applied elsewhere might not effectively manage or solve another class of problems – i.e. it might even make those worse off. This means that they are largely conducting research in which its lessons are "problem contingent". Any effort to generate lessons beyond a class of problems that is offered as a way to think about, and manage, all problems, is evidence that a scholarly analysis is not in this category. More specifically, any effort that concludes it is not "feasible" or "rational" to address a particular problem at hand that draws on a pre-established theory or framework that that was developed to analyse any and all problem is outside of this sorting category.

This elaboration leads to the identification of a more nuanced table of which human and AI coders must be aware. Coders will note that I have also changed the Y axis label to "problem contingent" which I treat as synonymous as "problem solving" dominates, but also more useful for coding purposes. I keep the broader term in the first table as it is more accessible to the uninitiated coder, while bringing in the more specific term once they have read these first few overview pages.

Table 2: The Four Problem Types- Expanded Distinctions

		Utility dominates? Does the analysis, conclusions, and theories treat individuals, organizations and states as largely self-interested, satisfaction driven entities that seek some kind "utility enhancing" outcome?

		YES	NO
Problem contingent? Is the analysis, conclusions, and theories generated from, and narrowed to, understanding, improving, or solving, a clearly specified "on the ground" problem or class of problems?	**YES**	**TYPE 1** **utility/ contingent**	**TYPE 4** **not utility/ contingent**
	NO	**TYPE 2** **utility/** **not problem contingent (universal)**	**TYPE 3** **not utility/** **not problem contingent (universal)**

In sum, at the risk of repetition, let me be clear: these frameworks offer ways to deliberate over ***not only how, but whether, a specific problem can be effectively managed given a broader set of considerations beyond the problem*** itself.

c) Examples of not problem contingent (universal approaches)

As I discuss below I refer to the major strand of Optimization school of sustainability thought and practice known as "cost-benefit analysis" (CBA) as a leading example of this approach. This is because CBA frameworks assesses whether or not a policy intervention governing a particular problem will be able to produce, or not, aggregate utility benefits. Under this approach, if policy designers cannot find an intervention to address a problem at hand that will enhance overall utility (i.e. possible policy interventions are all deemed to undermine aggregate utility) then the "rational" answer from this CBA approach is not to act.

There are, to be sure, many other strands of the Optimization school. For example, within international relations, both neo-realist and neo-liberal international relations scholars seek to generate theories of state preference, including longstanding debates about whether states maximize their preferences owing to absolute or relative gains. What is important for the 4PT framework is not their debates about underlying motivations that shape a given state's "self-interest" preferences, but that both neo-realist and neo-liberal institutionalist scholars develop their theories and frameworks to explain ***state behaviour in general*** – i.e. they seek universal understandings of utility chasing state preference that explain, but are not derived from, their choices about how to manage a specified problem. That is, they develop and apply theories of state preferences that apply beyond a particular problem, but are meant to explain state behaviour over a particular problem. To repeat, their project is to develop universal theories of state action that are not derived from, or limited to, any specific problem or issue states happen to be deliberating over. These theories of state action also inform recommendations from these scholar about how states might act go address a specific problem in order to better achieve their pre-existing self-interested preferences (i.e. how to "optimize" outcomes given existing understandings of state preferences). This is why, as I discuss below, so many IR scholars make the case for addressing the global climate crisis based on the argument that doing so will advance state preferences, rather than asking whether a theory of state preferences is the most useful way to think about, and manage, particular kinds of problems. This same dilemma facing cost-benefit scholars who so believe in the utility enhancing motivations that they constantly argue, and work so hard to demonstrate – even despite overwhelming evidence to the contrary – that an emphasis on

the utility undermining effects of the climate crisis is the most useful way to conceive of, and develop policy options for, averting global warming.

As I discuss below, strands within the "Compromise School" of sustainability thought and practice are also replete with a range of efforts to develop universal theories of stakeholder engagement, legitimacy formation, and conflict resolution that, while rejecting utility chasing theories, nonetheless also develop frameworks for understanding motivations of individuals, organizations and states, that are also developed to understand, and prescribe solutions for, a range of problems that may come across the desk of policy officials. For example, there is one strand of the Compromise school known as "alternative dispute resolution" ADR that offers a way for stakeholder to agree on litigation avoiding natural resources management agreements that draws on universal theories of individual and organization behaviour as a way foster a compromise agreement among disparate interests. While drawing on theories that extend far beyond, and reject, utility motivations as far to narrow an approach to human behaviour, they share the same mission in developing university theories as a way to generate an understanding about not only how, but whether, the "on the ground" issue, or set of issues, can be resolved given the overall focus on conflict resolution rather than solving and effectively managing, the "on the ground" problem that spark the ADR in the first place.

Another stand of the compromise school has been nurtured by those studying philosophy which seeks to generate a set of norms and protocols for making difficult decisions – such as whether to save someone stopped on the train tracks that might risk the lives of many on the train. In these cases, pre-existing approaches for developing normative theory based protocols for doing so – not based on utility per se but on claims to higher level "ethics" are turned to as a way to have society and policy makers thinks about managing difficult questions – from species extinctions that results from enhancing economic growth – to saving lives during a pandemic when the isolationist policies for doing say may undermine children's cognitive development. In these cases what is important for the 4PT framework is that the adjudicating analysis that result are meant to help policy makers and society make decisions to address any problem that might come along, and are not derived, narrowed to, or contingent, on a particular class of problems.

d) Examples of problem contingent approaches (not universal)

In contrast, and as I elaborate below, members of Commons and Prioritization schools narrowing that adjudicating frameworks to only address, and are indeed derived from, a particular "on the ground" problem or class of problems – whether it be for utility enhancing (Commons) or non-utility (Prioritization) motivations. There are many strands within each of these schools that reinforce "problem contingent" rather than non-problem contingent universal (i.e.. meant to be applied to any problems) frameworks. As I discuss below the most famous utility seeking, problem contingent strand within sustainability studies are those seeking to understand, and effectively manage, "tragedies of the commons" that, if not well managed, produce avoidable resource collapse. Here, scholars seek to understand key features of the problem – such as Ostrom's fundamental point that some commons tragedies – such as fish that swim - cannot be served by "excluding" access – and hence require different solutions to effectively manage them. The point for the "problem contingent" classification here is that scholars like Ostrom and her students who focus on "non-excludable" tragedies, only develop theories and approaches based on these key features. That is, they do not research, and have little to say, problems that do not share these key features.

There are many likeminded strands about developing problem contingent approaches by those who do not share the Common's utility orientation, but do share is focus on how to effectively manage specified "on the ground" problems – i.e. they eschew considerations about "whether" a specified problem can be solved, to deliberating over how it can solved or effectively managed. For example, Tribe's work on species extinctions rejected analytical universal approaches offered by extant "policy sciences"

framework, to show how any consideration to saving a particular species must begin, and end, based on the knowledge about what kind of conservation efforts is required to maintain the viability of that species. While Tribe and others like him turned to debates in philosophy and the study of "ethics", he rejected as problematic any effort to address species extinctions that might produce an outcome that caused species extinctions. Hence universal adjudicating frames about particular protocols for assessing policy dilemmas – based on universal theories of "ethics", was in fact, highly unethical. Others in this line of thinking within international relations have pointed to the "on the ground" problem of slavery. For this group, like species extinctions, the only way to eradicate slavery to develop policies and interventions based on achieving no-slavery. The idea of applying cost-benefit analysis, or alternative dispute resolution, to assessing not only how, but whether, slavery can be eradicated is anathema to this strands of the prioritization school because it opens the door to consider policy options in which slavery might continue. Hence for these scholars, adjudicating frameworks are always derived from knowledge to manage a specified problems (or class of problems with similar structural features and attributes, such as Ostrom's non-excludability tragedies). These distinctions have led Cashore to elaborate four categories to include these broader distinctions (Table 2) in which we can now further narrow "problem solving" to "problem contingent" which more accurately labels theories and frameworks and course of action that are never meant to be universal, but only applicable to a class of problems

e) Examples of the resulting Four Problem Types (4PT)

Now that we have covered how the 4PT framework defines utility and problem solving distinctions, and reviewed examples of universal versus problem contingent approaches, we are now in a position to identify examples of approaches classified within each: Type 1 (utility, problem contingent); Type 2 (utility, not problem contingent); Type 3 (not utility, not problem contingent); Type 4 (not utility, problem contingent). This section will, out of necessity, repeat some of the examples above. This is by design: repetition of examples will provide clarity when the same ones are used to distinguish different parts of the classification framework. In addition, this part is important since the coding is focused on the resulting four distinctions – so understanding how the above review figures for making decision about where to place a range of literatures is greatly facilitated by giving these concrete examples. This review, in turn, paves the way for a broader discussion of four key schools of sustainability that reinforce, but are also much broader, than the 4PTs. While broader, inclusion of discussion of the schools is expected to generate knowledge that will facilitate and enhance and generate a nuanced approach to coding the 4PTs.

i. Type 1: utility dominates, problem contingent

Type 1 then sorts those that are motivated by utility chasing orientations, but narrow their focus to a specified "on the ground" problem that is seen as undermining utility enhancement. Those scholars that are sorted into this category seek to generate analyses and ideas for intervening that improve utility outcomes for that problem. These approaches target the problem or class of problems they seek to ameliorate. Their interest is not in developing universal theories to address any and all sustainability problems. This means that they generate recommendations and analyses based on their problem or class of problems in isolation from other problems that might have different features. And they certainly would never apply a theoretical or analytical framework to discuss whether the problem should be solved.

There are countless examples of this kind of orientation. For example, I have a colleague who studies the issue of taxis idling looking for riders in downtown Singapore when it appears that they would make more income (utility) by driving to the airport. This problem then can be solved by reducing "information asymmetry" though some kind of intervention – such as advertising or sharing this knowledge, or through some kind of regulation reducing an ability to idle. In this case, interventions are adjudicated only for their ability to manage that particular problem - regardless as to the possible effects of the "effective

policy intervention" in reducing overall aggregate utility – such as those case in which limited government resources are better placed on managing another problem that would lead to higher utility.

As I discussed above and elaborate below Type 1 sorting would also include a large group of scholars and practitioners working on "collective action theory" in general, and its application to sustainability through the "tragedy of the commons metaphor" in particular. This is an important distinction. Collective Action theorists are motivate by eliminating utility undermining outcomes such as rubber necking on highways, hording of toilet paper during pandemics, or over harvesting that leads to collapse of fish stocks or water supplies. These approaches, which draw on the classic "prisoners dilemma" in which, in the absence of an ability to develop and comply with, a collective agreements, two prisoners in different cells both have a "self-interest" to confess because if they do not, but their partner in crime does, they will be handed a much higher and longer prison sentence. This "confessing/confessing" strategy is considered to the most effective utility enhancing approach given no collective agreement (an outcome known as the "Nash equilibrium) but it is not the most utility enhancing solution. The most utility enhancing solution is for both not to confess, as this will lead to lower prison sentences overall. In this case, the issue then is how to build and design collective action institutions to allow for both not to confess. In the sustainability world, the issue then is how to build and design collective action institutions such that harvesting of the resource in question is less than or equal to the biological rate or reproduction.

In all these cases, analyses are focused on human satisfactions, but only around the specific problem in question. As I discussed above and in more detail below, Elinor Ostrom's path to earning the Nobel prize in economics was sparked by her recognition that for some utility undermining tragedies, highlighted by fish, excludability was not possible. This mean that traditional interventions to solve this utility undermining commons tragedies that turned to private property or state ownership, and that assumed excludability was possible, were ill suited to addressing those utility undermining resource depletion tragedies. Ostrom then turned to assessing how to effectively manage these "common pool resources" (CPRs) which she posited were better addressed through local resource governing innovations.

The point for coders here is that all of these efforts paid attention to the key features of the problem at hand – specifically those contextual and contingent factors that explained the particular utility undermining outcome for the problem at hand. Collective action theorists in general, and commons tragedy scholars in particular, only generalize based on the specific problem at hand, or a class of problems such as those commons tragedies in which excludability is not possible. They would never therefore apply their lessons to managing all problems nor bringing in an assessment as to whether the problem can be solved. They only look at the how to do so, not whether.

ii. Type 2: utility dominates, not problem contingent (universal)

In contrast, **Type 2** literatures share an interest in understanding and championing utility seeking preferences, but they reject a focus that only includes attention to the specific problem at hand. The logic behind this, whether CBA that targets and aggregates individual preferences, or IR theories that target and aggregate state preferences, is that that group believes that it is unwise to assess policy options for ameliorating a specified problem without considering, and finding a way to incorporate, the result of this decision on other kinds of problems. Their logic is clear: if solving or effectively managing one specific kind of utility undermining activity – such as say, over harvesting timber – results in a set of policy and governance decisions that might preclude more lucrative activities – such as say the introduction of copper mining and its promise of increase wages – then policy officials have undermined their responsibilities in managing for the greatest amount of utility with which to benefit the welfare of citizens.

This category also captures the IR scholars above who also seek to generalize about the processes through which states do, or should, maximizing their preferences – often in the name of the "national interest"- that are then applied to assess how – and sometimes even whether – the problem at hand can be effectively managed given state preferences. Hence those analysts that turn to state preferences as a way to think about how to manage and develop policy response – whether it be, as discussed below, Nordhaus' reasoning that optimal state responses to the utility undermining impacts of the climate crisis will produce a 3.2 degree world – in contrast to the 1.5 outcome a problem contingent approach of the IPCC. It also applied, as discussed below, to neo-liberal institutionalist Keohane whose climate research turns to his theory of state preferences for policy solutions, rather than interrogating the contingent nature of the climate crisis itself.

Hence whether CBA, or conflict resolution or a host of other likeminded approaches, Type 4 classifies those analytical approaches that are not offered as contingent on solving a particular problem but as a way to adjudicate the problem at hand by apply a universal way to think about, measure, and identify, and project policy interventions. Hence, these approach there expands consideration, and offering a framework for doing so, about whether the problem at hand **can, or should,** be effectively managed based on some kind of approach to preference formation.

iii. Type 3: utility does not dominate, not problem contingent (universal)

There are many diverse approaches that are nonetheless united by their **Type 3 status**: i.e. analytical frames for understanding, explaining and prescribing policy actions that generate protocols for doing so that eschew utility motivations as a dominant force or outcome, and instead reflect on a diverse set of motivations with which to understand, and prescribe courses of action. These approaches are united by rejecting the treatment of non-utility outcomes as "irrational" and instead wonder how to understand and incorporate cultural values, social practices, environmental norms, that produce some kind of way to both understand the world and also provide the basis for recommendations about how to include a diverse set of ideas, organizations that generate legitimate process and wide accepted outcomes. As reviewed above when discussing universal approaches, while they reject utility as a dominant motivation, they share the idea that pre-existing theories and frameworks can be applied to adjudicate not only how, but whether, a particular problem can be solved. In other words, with these approaches – whether originating from a particular set of ethical frames in philosophy, sociology, geography or critical theory, one never knows for sure whether applying a particular one of these approaches to a particular problem will result in the problem at hand being solved or effectively managed. In other words, in contrast to the teleological orientation of Types 1 and 4, Type 3 approaches work from a preexisting framework – or develop a framework from a case that is meant to apply to any chase – rather than developing a framework that only pertains to that problem or class of problems.

iv. Type 4: utility does not dominate, problem contingent

For these reasons **Type 4** frameworks are distinguished, like Type 1, on their ability to help understand, and ameliorate, a specified problem from the Paris Accord's 1.5/2 degree objectives, to a particular species being threatened with extinction. However unlike Type 1 reinforcing approaches, Type 4 problems not only reject utility as the guiding force, they usually find that when those chasing utility outcomes are effective, they usually cause, or exacerbate, a range of Type 4 problems. The classic example comes to us from the identification of two distinct problems that are outside of the sustainability challenges to which Cashore devoted his empirical work. The first problem is that of slavery – i.e. the act of one human having property rights over another human being – i.e. "owning" them. A range of scholarly analyses assessing this problem have found that the only way to ensure slavery is dramatically

reduced or eradicated is to begin, and end, with the required outcome: i.e. no one, for any reason, is allowed to own another person. Many apply relevant frameworks have not only found that (Type 1 and 2) utility chasing motivations was behind slavery in the first place, but that application of Type 3, 2 frames permits deliberations over whether slavery ought to be considered – which, by definition, undermines an ability to achieve the desired outcome. The second example is anti-colonial norms in which reducing or averting an attempt by one country to take over another can only be achieved by teleological approaches that beginning, and end, with the absence of colonialism. Scholarly analysis reinforcing Type 4 hence would reject neo-liberal and neo-realist IR theories as permitting an understanding of state preferences that could also consider colonial type behaviours, rather than assessing mechanisms for ensuring these possibilities are never considered. As I discuss below, a number of analytical frames within environmental studies are consistent with Type 4 reasoning.

f) The Four Schools of Sustainability Policy and Governance Thought and Practice

Cashore's abductive application of the 4PT framework to, with, and from, a systematic review of sustainability policy and governance scholarship led him to identify four distinct "schools of thought and practice" that reinforced each problem type. Cashore's treatment of "schools of thought and practice" are treated broadly to cover key themes: Commons (Type 1 reinforcing); Optimization (Type 2 reinforcing); Compromise (Type 3 reinforcing) and Prioritization (Type 4 reinforcing). This broad treatment that there can, and are a number of unique strands – such as "cost benefit analysis" CBA and neo-realist and neo-liberal IR theories all falling under the "Optimization school". There is already much work on these strands that Cashore's categories help place, rather replace. The point here is that attention to the underlying aspects of these schools – one degree of abstraction up from the strands – allows for greater understanding of often unconscious biases of "thought and practice" that unite these strands. There are, to be sure, other schools of thought and practice that reinforce each Type – whether exploring the environment or for that matter public health or education or social problems. Cashore selected these four schools of thought and practice because they, arguably, represent the key forces of unconscious and conscious bias that have shaped, and changed, how we trains student and practitioners to view the world, bias empirical assessments and solve some problems that make other worse.

Table 3: The 4PT Framework and Four Reinforcing Schools of Sustainability Policy and Governance

		Utility dominates? Does the analysis, conclusions, and theories treat individuals, organizations and states as largely self-interested, satisfaction driven entities that seek some kind "utility enhancing" outcome?	
		YES	**NO**
On the ground problem? Is the analysis, conclusions, and theories generated from, and narrowed to, understanding, improving, or solving, a clearly specified "on the ground" problem or class of problems?		**TYPE 1 (Commons)**	**TYPE 4 (Prioritization)**
	NO	**TYPE 2 (Optimization)**	**TYPE 3 (Compromise)**

Source: Created by Cashore, first published in table format in Cashore and Bernstein (2022)

To identify these schools of sustainability thought and practice, and leading strands within them Cashore disentangled began by first reviewing their approach to **utility**/non-utility and **problem contingent**/non-problem contingent (universal) orientations and then added a number of other dimensions that shed

further light not only on the schools, but how these dimensions work to reinforce each of the four types. **Solution orientation** reviews the kinds of preferred policy solutions or interventions that the school, for various reasons, tends to favour over others. **Ethical motivations** captures the underlying purpose behind the emergence of the school in the first place that are explicit in some cases but implicit in others; **dominant disciplines** generates propositions about which ones seem more likely to champion this school over others; the **dominant methodologies** that result (especially concerning the quantitative and qualitative divide); the resulting and sophisticated **theory building project**; their **approach to effectiveness**; hidden or uninterrogated **implicit biases**; and finally, their effects in thinking about, and addressing, **birds and moles**: i.e. their biases in finding synergies by attempts to "kill two birds with one stone" versus their biases in inadvertently causing other problems that I refer to, drawing on earlier collaborative work, as the "whack-a-mole" effect.

i. Commons (Type 1 reinforcing)

Problem contingent? Yes. The Commons school is focused on designing "collective action" institutions capable of avoiding "tragedies of the commons" in which overharvesting causes catastrophic resource collapse. It draws on decades of attention understand and resolve "collective action" dilemmas (Goldstein and Lenway 1989) in which owing to absent or ineffective governance institutions individual utility enhancing "rational" behaviour in the short term leads to utility undermining "irrational" behaviour in the long run for both individuals and the broader community in which they are situated. Collective action scholars apply their lens to understand and manage "on the ground" problems from "rubber necking" on highways that cause utility undermining traffic jams, to the hoarding of toilet paper during pandemics (Cashore and Bernstein 2020). The role of collective action theory in sustainability studies was made famous by Garret Hardin (1968) most famously labelled these as "tragedies of the commons" which turned attention to goods such as grasslands, fish and timber, that have "subtractability" as their key structural feature: i.e. their use by one individual takes away the ability of another to also use that resource. This interest led Commons school has applied the collective action metaphor most notably in sustainability studies to the case of avoiding, or ameliorating, uncoordinated "open access" situations (Araral 2014) in which overharvesting leads to resource collapse. For these reasons they thought carefully about the types of institutional arrangements most capable of excluding access.

Utility dominates? Yes. Overcoming these undesirable conditions has been reinforced by a number of game theoretic metaphors (Nash 1953), most notably the "Prisoners dilemma", in which two thieves being interrogated in different cells and unable to coordinate make the utility chasing "rational" choice, to confess to even though collective silence would have produced lower aggregate prison terms. Hence collective action theory in general, and its application to sustainability studies in particular, begins, and ends, with the championing of optimal utility outcomes

Key features of a specified problem: form the bedrock of this schools approach, determining every other decision and approach. Initially and following Garret Hardin's famous essay generated widespread interest in finding ways to overcome "subtractability": i.e. those cases in which the use of a resource undermines the ability of another to use it. Ostrom would make profound innovative contributions by turning to conceptualizations of specific on the ground problems that for– for either biophysical reasons (such as the ability of fish to swim long distances) or traditional community practices (Gibson, McKean, and Ostrom 2000) – as having features that made it impossible or difficult for policy makers to exclude access (Ostrom and Ostrom 1972).

Solution orientation: two overall approaches emerged initially within this school: either privatizing land to individuals (who would now have, it was posited, individual incentives to produce long term utility enhancing benefits), or to turn to coercive state central authority and associated enforcement mechanisms. Ostrom would make profound innovative contributions by noting that for– for either biophysical reasons

(such as the ability of fish to swim long distances) or traditional community practices (Gibson, McKean, and Ostrom 2000) – as having features that made it impossible or difficult for policy makers to exclude access (Ostrom 1990). Formal incorporation of "excludability" yielded not one but four ways to classify, and label, economically suboptimal resource depletion challenges: subtractable and excludable "private goods"; subtractable but not excludable common pool resources; non-subtractable and excludable "club goods", and non-subtractable, non-excludable "public goods". Although many of Ostrom's students self-label today as environmental governance scholars, there is no dispute that GTC's focus, and design principles, were explicitly derived from economic sustainability challenges (Ostrom 1990, 3). However, what is interesting, and long forgotten, is that although her work would largely reinforce Type 1 problems, she saw an equally compelling need to engage in similar attention to Type 4 problem structures for those seeking design insights for ameliorating the degradation of ecological and biological systems (Ostrom 1990, 1).

Table 1: XX
Conception of Resource Goods

		Excludability	
		Is the renewable resource in question easily "excludable"?	
		Yes	No
Subtractable? (Does using it reduce it?)	Yes	Type 1 private goods (e.g., timber)	Type 1 common pool goods (e.g., fish stocks)
	No	club goods (e.g., toll roads, professional associations)	public goods (e.g., air)

NOTE: Examples of goods in each cell are illustrative. Commons and sustainability scholars continue to differ on whether to treat trees, fish, and ecosystems as "private goods", "common pool resources" and, in some cases, as "public" or "club" goods. These debates can be traced back, in part, to different ways in which categories and corresponding solutions are constructed (Young 2007: 3-4). For example, following Ostrom's work in Governing the Commons, some view institutions, depending on their design, as causing or solving, a particular resource depletion tragedy (i.e. they constitute the "independent" variable). However, others view institutional arrangements, not simply biophysical features, as playing a role in making excludability difficult (i.e. they constitute part of the "dependent" variable). It follows that if a researcher deems institutions to have played a role in making excludability difficult, it is equally plausible, in contrast to GTC's inferences, to consider designing institutional innovations to make exclusion possible. This ambivalence in treating excludability as fixed and hence a common pool resource, or as changeable, and hence a "private good" is reflected in GTC's conclusions that resource quotas that exclude access are useful for managing, rather than converting, CPRs. Similar ambivalence is behind Ostrom's student Prakash and his colleague Potoski (2006), who flip GTC's treatment of "club goods" as a type of resource (i.e. the dependent variable) to instead, as a utility enhancing institutional innovation that might produce ancillary Type 4 environmental benefits (i.e. as an independent variable).

Source: Adapted from V and E Ostrom (1972), Prakash and Kollman (2004) and Gibson, Mckean and Ostrom (2000)

Ostrom also made it clear in GTC, and subsequently, that she believed that the tool of privatization, rather than coercive state control, was the most optimal for managing "strictly" private goods (Ostrom 2003, 259; Gibson, McKean, and Ostrom 2000, 7).[1] Her concern was that the widespread application of the privatization solution to include non-excludable CPR type commons tragedies drew on "special models that utilize extreme assumptions" (Ostrom 1990, 183) that could produce even more tragic outcomes than if nothing had been done at all (Ostrom 1990, 13-14) such as the expropriation of communal forests that had historically avoided resource collapse (Bartley et al. 2008; Gibson, McKean, and Ostrom 2000). GTC further posited that local communal governance institutions, including "groups of individuals", rather than "distant" central authorities, were best poised to avoid or reverse, many types of unsustainable CPR practices (Ostrom 1990, 21-23, 60-61, 183-185). While this approach helps account for today's "subsidiarity" metaphor among development scholars, similar inductive reasoning however has led the commons school, guided by the "polycentric" metaphor, to advocate for the "scaling up" of its design principles to ameliorate economically sub-optimal "global commons" challenges that cut across multiple jurisdictions.[1]

Ethical motivations: researchers are often motivated by an expressed moral duty to help avoid the devastating effects that the collapse of an economically beneficial resource wreaks on human beings, especially resource dependent communities. To do this the guiding moral concern is to overcome these suboptimal outcomes through the development of collective institutional arrangements. Ostrom championed the primacy of this school in her presidential address to the American Political Science Association, arguing that: "… the theory of collective action is the central subject of political science" (Ostrom 1998, 1).

Dominant disciplines: the discipline of economics forms the bedrock foundations for the Commons school, given its focus on utility enhancing management of natural resources. At the same time, especially following Ostrom, the focus on community governance as a preferred solution has drawn interest from anthropologists, and cousins in geography and sociological whose grounded theory methods provide necessary nuance for developing effective locally specific institutions .

Dominant methodologies*:* the search for theories to inform particular Type 1 resource depletion tragedies – helps explain the commons school embracing of game theoretic models, and quantitative methods required to build them. Qualitative ethnographic methods play a supporting role in so far as their insights help improve model building (Brondizio, Ostrom, and Young 2009; Araral 2014, 12; Andersson, Evans, and Richards 2009).

Theory building project: *commons scholars and general, and Ostrom in particular, are* clear that they are motivated to initiate "*better* policies" (Ostrom 1990, 29) for averting, or avoiding, depleting water, fish, timber, and other goods that contribute to human food systems, required "arranging for the supply of new institutions" (Ostrom 1990, 14) that must be placed on "how *best* to limit the use of natural resources so as to ensure their long-term *economic viability*" (Ostrom 1990, 1). Ostrom offered that getting the utility enhancing "institutions right" requiring identifying and granting "user rights", reducing "free-riding" risks and "commitment problems". She cautioned that successful efforts in "monitoring individual compliance with sets of rules" (Ostrom 1990, 27) would be "a difficult, time-consuming, conflict-invoking process" (Ostrom 1990, 14).

Birds and moles: following Ostrom's seminal contribution, commons school scholars are now keenly aware of, and think deeply about avoiding, the devastating internal "whack-a-mole" effects that can occur when treating all commons tragedies as excludable. Less attention has been placed on three external whack-a-mole effects that fall outside its moral, theoretical and conceptual foundations. First, successfully targeting a specific good could result in an economically suboptimal Type 2 outcome if converting the land to a different commodity or use – such as mineral extraction, manufacturing, real estate or even tourism – would have produced higher society wide aggregate economic utility. Second, while the Commons School certainly turned to the importance of local cultural practices for their role in fostering collective long-term Type 1 sustainability in the absence of formal institutions (Ostrom 1990, 149, 166; Sethi and Somanathan 1996), it does not formally incorporate into its model historical cultural traditions that fall outside of, or are degraded by, an emphasis on economic utility (Buck 1998). Third, and most important for our analysis, negative environmental effects can, and usually do, result from the solving a resource depletion tragedy, including, for example, the ecologically degrading effects of sustained yield logging and fishing on forest and marine ecosystems. To be sure, Commons scholars have certainly considered the impact of utility enhancing frameworks on environmental degradation. However, arguably owing to this school's deeply entrenched moral foundations, such reflexive exercises tend to advance causal beliefs that Type 1 and Type 4 problems are synergistic (Prakash and Gupta ; Costanza 2006).

While this school is sophisticated about what to do to solve a collective action challenge, it does not provide intellectual guidance about which problem, from sustained yields of timber, to exotic crops from soy to palm to eucalyptus, to fisheries to local water management, ought to be given attention. This means that if effectively addressed, the solution may result in synergies and moles that were not part of the inductive reasoning that led to managing the specified Type 1 problem at hand. For example, managing fisheries as a Type 1 problem may, depending on the reproduction logic of the resource in question, require maintaining the entire structure and function of the ecosystem itself—even though that is not the concern of Type 1 problem frames. Interestingly the main recommendations from the 2005 economics of water report (Ovink 2024)– that was produced to mirror the Stern review (2007)on the Type 2 economics of climate and Dasgupta review (2021)on the Type 2 economics of biodiversity essentially

concluded that if Type 1 water management across the world was to be conserved, the Type 4 ecological structure and function of forests and biodiversity needed to be maintained. However, the reverse can also be true, such as when successfully managing for the highest Type 1 sustained yields are found to undermine ecosystem system structure and function and/or render extinct particular species (Cashore and Howlett 2007). Likewise, the Commons school is largely silent on what to do when successfully designing institutional arrangements for managing a particular good in the long run might undermine an ability to convert the land to a more lucrative crop, such as palm oil or soy plantations, that would have provided even higher utility enhancing benefit (Cashore and Bernstein 2022).

Effectiveness: For all these reasons the Common's school is anchored in achieving desired policy objectives facing a clearly specified, and measurable, "on the ground" challenge whether it is about maintaining the long-term supply of utility enhancing fish for a poverty stricken community, timber management in 1000s of forest dependent communities across India, or long term supplies of clean water facing local urban areas, there is no success, and no sense of accomplishment, if the specified issue at hand becomes degraded or collapsed. This is important because its attention on revisiting theories and prescriptions if they don't cause the resource good to be maintained (rather than simply managing degradation more slowly) is not only a critically important aspect of this School for Sustainability scholarship policy and governance, its approach to "on the ground" effectiveness in managing the particular Type 1 problem at hand, can, and must, be emulated by Type 4 problems that consumer the interest, and attention, of the Prioritization school. In sum, this school's problem orientation is decidedly not focused on ecological, but economic outcomes. As Ostrom made clear, "The issue in this case—and many others—is how best to limit the use of natural resources so as to ensure their long-term ***economic*** viability" (Ostrom 1990: 1).

ii. **Optimization (Type 2 reinforcing)**

Problem contingent? No. Those who belong to this school of sustainability thought and practice do not justify their analytical frames based on key features of a specified problem. Instead, they are are united by developing approaches to thinking about, researching, and identifying solutions for managing any kind of problem that might reach the societal or policy agendas. They are united by theories and approaches that do not derive from the specific features of a particular kind of problem, but for developing universal approaches to address not only how, but whether, a problem can and ought to be effectively managed. To be sure, the key features of any "on the ground" problem at hand are certainly not ignored, but they do not constrain or narrow the universal theory building project about how to understanding satisfaction utility enhancing behaviours on the part of individuals, organizations or states. They may use an on the ground case to generate a universal theory, but the general theory will be applies beyond the case to to explain behaviour in general, and to identity and narrow policy solutions

Utility dominates? Yes. Whereas the commons school champions a utility maximizing outcome to manage a clearly specified collective action problem, the optimization school advances the moral belief that any problem at hand must only be solved if solutions enhance the utility of individuals, organizations and states at the aggregate level (i.e. they can only effectively manage a particular problem at hand if doing so enhances satisfactions overall. If satisfactions are undermined overall by solving one problem in isolation, they develop implicit and explicit frames and methodologies that reinforce a morally inappropriate do solve a problem at hand if it undermines overall aggregate utility of individuals, organizations and states.

The major strand dominating sustainability policy and governance within this school are those who practice some kind of CBA using neo-classical or behavioural techniques nurtured within the economics discipline. If a solution cannot be found to the specific issue at hand that doesn't result in the undermining of overall aggregate utility- what is also referred to as "social welfare", then the "rational" answer is that

it cannot be solved. It is for these reasons that this school often advances a "partial" solution that is not in line with the problem at hand but does do something. Hence the Optimization school shares an interest with the Commons school in advancing some kind of utility or satisfaction for humans and their organization, but differs in that it does not just address how to solve a problem at hand, but whether it is "rational" to do so.

Solution orientation: For these reasons the solution is around maximizing satisfactions over all, which will determine not only how, but whether the problem at hand can be addressed.

Ethical motivations: Within sustainability studies, this school's moral framework provided the dominant rationale for the creation of forestry, agriculture, and resource schools in Europe and North America that were motivated by utilitarian principles[3] of "the greatest good for the greatest number in the long run" (Pinchot 1987), as well as most post-WWII development agencies, including the World Bank. Of course, these are deeply held. To be sure, specific strands have all developed deeply ethical justifications. For example. The economic optimization strand of this school advances the moral belief that the quest for policy solutions that enhance aggregate economic utility is the best outcome for society as a whole (Kenny 2011; Sen 1979; Luke 2009). In contrast the international relations strand focuses on understanding why states cooperate, and based on that answer, to identifying, and helping create, the most "optimal" institutions to achieving the highest level of preferences. To be sure, the massive debate within this strand about whether relative or absolute gains motivate states to engage in international cooperation reinforce the universalist nature of this project that is disconnected from, but meant to be applied to, any policy that might come on the geo-political agenda.

Dominant disciplines: Economics, rational and public choice political scientists, international relations scholars including neo-realists (emphasizing relative gains) and neo-liberal institutionalist (emphasizing absolute gains).

Dominant methodologies: The nature of this school's quest for a universal theory to explain the empirical world they are interested in, from aggregate economic utility to optimal preferencing building state cooperation, means that quantitative methods are, and must be, dominant. These include econometrics modelling and game theoretic analyses. The most dominant is "cost-benefit" analysis (Adler and Posner 2009; Arrow et al. 1996) in which disparate social, environmental, cultural and economic goals are given comparable economic utility values. The field of environmental economics advances this method based on their causal belief that the rational and feasible way to address environmental challenges is to convert them into economic values (Stavins 1995; Sachs et al. 2020; Thomas and Chindarkar 2019). This approach, which underpins the US Environmental Protection Agency's policy to value each statistical human life at $9.47 million USD (Environmental Protection Agency 2020) involves modelling across a range of outcomes, as well as discounting future economic impacts, to identify the "best" or most "rational" policy options. Internal debates – including whether policies must be avoided that increase economic inequality (Kaldor 1939) – whether measured as widening gaps in relative gains (Piketty 2015) or Pareto losses (Awan 2013) - reinforce the school's overarching emphasis on utility enhancing orientations by narrowing consideration of inequality to economic measures. Other methods for giving utility enhancing equivalents include the production of quantitative surveys of consumers and the public's "willingness to pay" so that concerns about species extinctions, wilderness, and the catastrophic ecological effects of climate change can be converted into economic values (Turner et al. 2003)

Theory building project: arguably the most dominant of these strands today is economics which advances the causal belief that the rational and feasible way to address environmental challenges is to convert them into economic values (Nordhaus 1991; Stavins 1995; Sachs et al. 2020; Thomas and Chindarkar 2019). Drawing explicitly or implicitly on welfare economics,[1] that the ability to solve any kind of problem is conditional upon finding policy solutions that enhance aggregate economic utility for society as a whole

(Kenny 2011; Sen 1979; Luke 2009).

This school of thought and practice also includes international relations sustainability scholarship on cooperation and rational institutional whose sophisticated analytical frameworks are designed to extend beyond, and are not derived from, the structural features of the environmental problem at hand. Rather, they generate their underlying theories from attempts to understand state cooperation generally that apply to any and all problems and from that, identify possible solutions to the problem at hand, such as climate change (e.g.Keohane 2016; e.g.Keohane and Victor 2016; Ovodenko and Keohane 2012; Koremenos 2016). To be clear, this does not mean that these scholars do not care about identifying possible solutions to the Type 4 environmental climate, biodiversity and forests crises – many of them do. Rather, their ability to do so in not derived from the key features of the problem at hand, but their theories of international cooperation.[4] It also applies to rational choice strands within political science who seek to understand ways to construct and maintain governance institutions capable of enhancing aggregate utility

Effectiveness: This school of thought therefore is not about achieving an environmental outcome qua and environmental outcome – such as say, doing what biologists say is required for averting a species from becoming extinct – but instead by whether this problem was handled in a way that maintain aggregate utility or some other equivalent, such a enhancing state preferences, whether based on absolute or relative gains. Hence, and given this school's ethical beliefs, it turns to empirical measures of aggregate utility, and corresponding asserted benefits for society that will accrue including poverty alleviation, directly and indirectly. Direct benefits include higher levels of employment and income. Indirect benefits include enhanced resources for health care, schools, and other social programs that can produce and improve the human conditions (Rawls and Laband 2004). This approach of this school has been credited with generating effective policy designs that have lifted millions out of poverty (Hasan 2021).

Implicit biases: Adherents to the optimization school relay on utility maximization to assess not only how, but whether, it is rational to any particular problem that comes across the desk of a policy official, including climate and biodiversity.

Birds and moles: For these reasons the Optimization school is devoted to bringing in as many problems as possible into their research and applied analysis so that policy officials do not inadvertently make a "suboptimal" choice owing to lack of information. This allows them to focus both on "positive" (birds) and "negative" (moles) externalities. This schools' approach certainly can, and does, lead to multiple tentacles associated with utility maximization such as greater community stability, lower levels of violence, and greater access to education and health care that often result from increasing income- even though the extent to which these will occurs will always be uncertain. The CBA strand addresses moles as unintended "externalities" that must be measured by giving these outcomes utility undermining values – usually through "willingness to pay" surveys – and then incorporating this into their models as a way to "internalizing" these "externalities". This approach is reinforced by their ethical foundations, and causal beliefs, that valuing nature economically *will* provide ancillary environmental benefits (Sell et al. 2007). This morality is also behind the "polluter *pays*" metaphor (Díaz-Reviriego, Turnhout, and Beck 2019; OECD 1975, 1994) which holds that firms can chose to stop an environmentally degrading behaviour if the (economic) costs to the firm of continuing are too high, or it can continue polluting if the costs are deemed low and behind their turning to policies that encourage "payment for ecosystem *services*" (PES) (Sell et al. 2007).

At the same time this school's approach to address moles by giving them utility equivalents, can, and does, create new Type 1, 2 and 4 moles that neo-classical and behaviour economics in general, and the CBA strand in particular, rarely think about, let alone address . In a classic and disturbing example to members of the Prioritization school, (Type 2 reinforcing) economist William Nordhaus' has concluded, based on his sophisticated modelling of the economics of climate degradation[5], conclusion that that the

most optimal, or "economically rational" future are world temperatures at 3.2°C warming above natural levels (Nordhaus 2017) despite (Type 4 reinforcing) scientific evidence produce by the world's leading scientists that 1.5/2°C is the maximum warming possible without risking catastrophic ecological changes (Cashore and Bernstein 2022; IPCC 2018). Hence, in Nordhaus' case even though he is focused on the climate crisis he is not problem contingent: i.e. he did not develop a theory and approach to solving the climate crisis at 1.5 or 2 °C. Rather he applied a pre-existing framework focused on maximizing aggregate utility that determines both how, and whether, a problem can be "rationally" solved. Keohane, chasing a theory of state cooperation based on absolute gains, takes the same overall ontological approach, albeit applying his own theories and methods.

Hence, while this school is sophisticated at internalizing a range of outcomes into utility enhancing models – it pays no attention to, nor does it accept as addressable (Rawls and Laband 2004), moles that this school creates. Most notably, its emphasis on economic growth can, through resulting industrialization processes, result in the homogenization of previously diverse local cultures, while associated increases in consumption can significantly stress the earth's environmental "carrying capacity." Similarly, its emphasis on existing "willingness to pay" calculations to convert environmental values into equivalent economic utility comparators, undermines reflexive considerations about the role of norm generation that may shift how humans conceive of ecological systems – such as viewing them not as providing services to humans to but as places worth of protection in their own right: i.e., conserving ecological systems qua ecological systems. In the most extreme, some rational choice political scientists have gone so far as to treat Type 1 and 2 human satisfying outcomes not only as synergistic, but as synonymous, with Type 4 environmental concerns (e.g. Lomborg 2001).

iii. **Compromise (Type 3 reinforcing)**

Problem contingent? No. The compromise school seeks to generate legitimate and trustworthy processes through which multiple perspectives and concerns are included when deliberating not only how, but whether, to ameliorate any given problem at hand. Like the optimization school the quest is not to develop a stakeholder process based on the specific features of the problem at hand and working backwards to find an effective policy solution, but rather, to apply a pre-existing approach to deliberations that the public, stakeholders, and policy makers deem to be appropriate.

Utility dominates? No. This problem framing have been prominent at the international level since 1987 Brundtland Commission (WCED 1987), whose championing of a "three legged stool" of sustainability, as an *objective* way to achieve *sustainable* outcomes, permitted a bridge conversation with, but then sparked an "outside" critique of, the economic optimization school. At first, the Brundtland report articulated a now widely diffused a Type 2 anthropocentric moral imperative that sustainability required "meeting the needs of the current generation without compromising the ability of future generations to meet their own needs". However, this framing would subtly but powerfully confront the economic optimization strand's narrowing of environmental and social problems to their utility enhancing benefits (not to mention that most in this strand, until Stern, accepted as objective, the subjective belief that longer term Type 2 outcomes must be "discounted" compared to immediate ones). These tensions led to a second metaphor post Brundtland in which a conversation with the Prioritization school (Wang 2004) led to the incorporation the impacts of humans on the natural environment on the belief that sustainability must incorporate, but treat as distinct, environmental, social and economic goals through the "three legged" stool metaphor.

Solution orientation: the Compromise school incorporates, but expands, an emphasis on utility to include, in their own right, environmental, social, and cultural goals (Cashore and Bernstein 2022). Its ethical foundations, highlighted by the Brundtland Report (WCED 1987) are aimed at promoting balance

among competing goals and interests through principles of inclusiveness, transparency, accountability, and empowerment (Cashore and Nathan 2020).

Ethical motivations: the compromise school's moral philosophy rests on the belief that when a specific "on the ground" problem comes on the policy agenda it is morally appropriate and fair to engage a range of ideas, interests and stakeholders with diverse values and concerns, including marginalized and vulnerable communities, groups, and people. This school's moral foundations from a rich literature from pluralism to tri-partite corporatism that is devoted to understanding, and building "good governance" at multiple scales through enhancing legitimacy and trust (Dahl 1961; Rosenbluth and Shapiro 2018; Bodansky 1999; Habermas 1973; Scharpf 1999; March and Olsen 1998). Various stands of this school articulate there mission in specific ways, such as the Lasswellian "policy sciences" community united around the promotion of "human dignity" rather than economic utility (Farr, Hacker, and Kazee 2006). Within sustainability studies, these themes are reflected in a rich set of scholarship aimed at understanding, and assessing, how inclusionary processes and deliberative spaces might foster meaningful involvement of disempowered actors, generate new ideas (Dryzek 1990; Hoberg 1992; Pinkerton 1993), and perhaps improve distributional considerations (Aklin and Mildenberger 2017).

Dominant disciplines: include political science (especially strands on legitimacy, state-society interaction), sociology, psychology, anthropology, economics

Dominant methodologies: for this school require, out of necessity, mixed approaches including of quantitative and qualitative analysis. Special attention has been placed on insights for generating consensus among environment and development "advocacy coalitions" (Sabatier 1988). Applied policy analyses turns to some type of "multi-goal" policy analysis (Weimer and Vining 1992) and stakeholder deliberations to achieve consensus and/or "dispute resolution." Significant research is conducted to assess how representatives of different interests or values might be incorporated that avoids the tendency to drift towards Type 2 (Raymond and DeNardis 2015; Clémençon 2012).

Those scholars in the applied space include turn to "multi-disciplinary" or "interdisciplinary" knowledge generation owing to causal and moral belief that integrating a diversity of perspectives *will* yield the most legitimate and effective policy responses (Nilsson and Weitz 2019; Saez and Requena 2007; Clark and Wallace 2011). This strand of the compromise school includes an important network interdisciplinary collaborations with natural scientists – including some trained in Ostrom's Indiana workshop – aimed at promoting Bruntdlandesque morality as a *scientific* endeavor (Schmitz 2018; Reed et al. 2019; Matson, Clark, and Andersson 2016; Glaser and Bates 2011). These efforts culminated when the US National Academy of Sciences - whose members strongly bias, and reward, quantitative research- defined and formalized the compromise school's subjective conception of "sustainability science" as "…the interactions between natural and social systems, and with how those interactions affect the challenge of sustainability: meeting the needs of present and future generations while substantially reducing poverty and conserving the planet's life support systems" (Kates 2011, 19449).

This, in turn, would pave the way for many sustainability scholars to present their subjective Type 3 cognitive frames as objective by making universalist claims that their scientific research *found* that sustainability solutions *required* compromise be reached among disparate groups (Jacobsen and Linnell 2016; Grove and Forster 2011; Bryant and Jackson 1999; Baker et al. 2009). Practitioners and policy officials would draw on these conclusions to justify as "scientific" their quest to elaborate the "foundations" (Starik 1995) of "multiple-use" land management (Shands 1988).

These moral frames shape research agendas, including a plethora of mostly large-N quantitative studies in scientific journals including the Proceedings of the National Academy of Sciences, but they would be drawn upon by business interests and associations, such as the World Business Council on Sustainable

Development (WBCSD), to champion and fund research and teaching by the compromise and economic optimization schools.

They would also be drawn on, as the owl case reviewed in Part III illustrates, to sparked, and reinforced, a plethora of research into the conditions through multi-stakeholder processes might end the "war in the woods" by collectively defining, and practice "sustainable" resources and forest management (Davis 1992; Breyman 1993; Dietrich 1993; Gup 1990; Hogarth 1991) which was identified as in opposition to "elitist" Type 4 species conservation policies that were bureaucratic, anti-community and, was sometimes asserted, anti-democratic.

A much more subtle, but arguably even more powerful strand of this school was articulated by US and global scholars who, following their concerns about conflicts over debates about land use among environmental activists, indigenous peoples, resource dependent communities, labourers, and businesses sought to understand better the processes through which respectful dialogue might lead to legitimate solutions aimed at avoiding conflict and producing consensus through compromise (see, for example, Halbert and Lee 1990; Wondolleck 1988; Crowfoot and Wondolleck 1991; Bacow and Wheeler 1984; Wilkinson and Anderson 1985; Bingham 1986; Pinkerton 1993; Armitage et al. 2009; Kelly and Alper 1995; Coglianese 1997; Beyers 2001; Brach et al. 2002; Koontz and Thomas 2006).

Theory building project: turns to understanding, or constructing, governance arenas that, through particular ways to foster deliberative decisions making will lead to them being evaluated as so appropriate that its authority is accepted, even if a particular decision (Scharpf 2000),at any one point is not consistent with what a particular groups or interest would have preferred (March and Olsen 2004). Researchers from this school have found, or posited, that depending on how they are constituted, inclusionary approaches can help foster political legitimacy (Bodansky 1999), public support, and improve policy making (Dryzek 2005; Eckersley 2004; Bernstein 2017; Eckersley 1992).
Some strands of this school emphasize how particular engagement might help achieve consensus outcomes (Risse 2000).

Effectiveness: Adherents to the Compromise school assess success as creating some kind of stakeholder inclusionary process that would be reuired to balance economic, environmental, social and cultural values in ways that produced evaluations of trust, legitimacy and appropriateness of the process (Overdevest and Zeitlin 2016).

Birds and moles: The result was an approach to whack-a-mole effects by striving to balance and compromise across economic, environmental and social goals. However, the school is generally silent on the risks of several external moles. For example, as we will see in Chapter Three, Type 3 stakeholder engagement at domestic and global levels often produce "classic incremental" changes that rarely veer far, and often revert back to, the previous equilibrium. This is often owing to the role of powerful business interests in limiting significant change, while reducing the influence of those whose main objective is to champion environmental values and outcomes (Coglianese 1997). For example, powerful C02 emitting business interests who "…consider climate impacts as harmful but discount them, or consider the costs of climate action as excessive" have been found to be "veto-players" in international climate deliberations (Mitchell and Carpenter 2019, 5). Even when well designed to minimize the role of powerful interests, the compromise school's emphasis on legitimate processes and balance can lead to disastrous consequences for the problem at hand. For example, application of Type 3 multi-stakeholder engagement processes to address overharvesting in the Newfoundland cod fishery a quarter century ago resulted in a compromise decision to allow a harvesting rate higher than what biologists, based on careful attention to the key features of problem at hand, had recommended was required to maintain long-term sustainability (Chase 2003; McKenna 1992). The result was that the Type 1 fisheries economy collapsed, further degrading the Type 4 marine ecology from commercial fishing had already caused (Berrill 1997).

To be sure, decisions from this school can produce moles and synergies. For instance, a deliberative approach to a range of interests can produce greater feelings of trust and inclusion that create "peace dividends" that extend far beyond a particular compromise process (Bunse and McAllister 2022). However, this school risks producing, and even supporting, solutions that are inconsistent with the science of what is required to address a particular "on the ground" problem (Cashore et al. 2019).

For instance, Victor (2015), thinking about the agreement rather than its effectiveness, adjudicated the Paris Accord as "transformative" for generating support while simultaneously declaring the 1.5/2°C objectives as "ridiculous. Similarly, the Canadian province of Newfoundland invoked a Type 3 reinforcing compromise dialogue to address biologists' warning that its cod fish were being overharvested. These transparent, trustworthy, and legitimate deliberations led to an agreement on catch levels that were *higher than* what biologists projected would be required to maintain long term sustained yields. The result was the catastrophic collapse of Type 1 g cod fishery, and further degrading those concerned about Type 4 ocean ecosystem degradation (Cashore and Bernstein 2022).

iv. Prioritization (Type 4 reinforcing)

Problem contingent? Yes. In contrast, the Prioritization school (Type 4 reinforcing) confronts and interrogates the specific nature of the problem at hand – much the way Ostrom did so for Type 1 probems with her attention to "excludability", this school does the same to understand how, not whether, a particular environmental problem can be solved. For example, my careful attention to this question with Ishani Mukherjee, Lahiru Wijedasa and Altaf Virani, to Peatland, found that a key features was that its key features meant that it was impossible to maintain with irrigation – which would ultimately produce resource collapse (Cashore et al. 2024). Likewise this classification is consistent with those seeking to avoid erosion of hillside streams which the science shows that if policies must conform to the key feature that any number of trees must not be removed (Sabatier and Jenkins-Smith 1993). Hence this school – whether looking at a particular species or the 1.5 degree imperative spends careful time on the nature of the problem, and works inductively, or teleologically, to uncover those key features that would need to be ascertained to effectively manage the problem.

Utility dominates? No. In fact, this school has documented overwhelming evidence corroborating Cashore's Chapter One, that almost all Type 4 problems almost always result from, or are exacerbated by, highly effective Types 3, 2 and 1 policy solutions (Sinden, Kysar, and Driesen 2009; Zeckhauser and Schaefer 1968; Ackerman and Heinzerling 2004).

Solution orientation: confronts "commensurability" biases (Tribe 1972) inherent in the approach of the Commons and Optimization schools to instead, advancing a "sequentialist" approach that ranks "on the ground" problems in order of importance to policy makers and society, rather than turning to cost-benefit or multi-goal policy analysis to make these choices for them (Cashore and Bernstein 2022).
a classic example concerns anti-slavery as a policy objective that trumps utility calculations or compromise solutions (Cashore and Bernstein 2022). Hence, the prioritization school's approach narrows solutions capable of achieving, rather than compromising away, the problem at hand through "fit for purpose" environmental solutions. For example, when it comes to climate they turn to logical causal reasoning by noting that failure to grant the climate and species extinction crises as type 4 challenges risks adopting policy and institutional solutions that that are inconsistent with, or gloss over ecological knowledge: i.e. producing Newfoundlandesque outcomes for the world's most important ecological challenges (IPCC 2018; IPBES 2019).

Ethical motivations: Its moral framework is guided by the belief that some problems are so important that they are incommensurable to others. It draws on rich philosophical and theoretical literatures which reflect on how to distinguish "first order" principles and problem (see, for example, Rawls 1971) from "second order" problems. Tribe (1972, 95) explicitly confronted the "ill fit for purpose" quest of generalizable approaches to policy analysis for first order problems (ibid). The classic example concerns the eradication of slavery in which the very application of economic optimization or compromise approaches as adjudicating frameworks turns analytic attention to assessing, empirically, the pros and cons of owning slaves, which automatically undermines the problem conception and outcome itself: i.e., that no one, for any reason, ought to be allowed to own another human being.

This school was drawn on to understand, and inform, the policy, legislative, and agency reforms during the "first wave of environmentalism" in the 1960s and 1970s (Carson 1962; Paehlke 1992), as well the emergence of international attention on the effects of human activity on the natural environment (Willetts 1996; Meadows et al. 1972). Tribe offered that leading contenders for Type 4 status, included "…vulnerable or 'fragile'" problems especially "…ecological balance, unspoiled wilderness, species diversity, and the like…[that are]…are intrinsically incommensurable, in at least some of their salient dimensions, with the human satisfactions…" (Tribe 1972, 96). To be sure, Tribe's point was not that all environmental challenges must be conceived of as Type 4 problems, but that policy analysis techniques should not undermine such a possibility:

> …it might well be the case that there is no species whose complete preservation should have lexical priority over all human goods; but it simply does not follow that no minimal level (or perhaps some "overall" level) of species survival or diversity should [not] receive lexical priority over at least some kinds of human benefits …

Tribe offered that it would not "…seem peculiar to insist…that a lower limit be established on especially cruel treatment of animals, whatever the [Type 2] economic gains this cruelty brings to persons in general" (Tribe 1972, 96; parenthesis added). To overcome these commensurability biases, Tribe advocated a "lexical" approach in which the central task of the policy maker is to identify the structural features of the "first order" problem at hand, and then, just Ostrom did for Type 1 commons challenges, inductively identify policies relevant institutional and policy solutions irrespective of "second order" effects (Tribe 1972, 95, 96, 99). Hence, the Prioritization school's ethical philosophy include the imperative to render explicit hidden conceptions inherent in all policy analyses, and then to consciously, rather than inadvertently, design policies consistent with preferred biases. For these reasons the inherent ethical philosophy behind the Prioritization school is that lower ranked problems—including moles cause by effectively managing higher ranked problems - can only be managed in ways that do not undermine lexical priorities.

Dominant disciplines: within sustainability studies, the prioritization school has strong roots going back to critiques by biologists, environmental activists, and environmental agencies who collectively posited, and empirically found, that an emphasis on (Type 1 or 2) economic utility tended to be correlated with the degradation of the structure and function of ecosystems and species extinctions (Taylor 1992). It also has strong roots within historically oriented political science, sociology and environmental studies and humanities who, by standing back and viewing trends over decades have been able to view "the forest for the trees" (or perhaps the lack of forests ☹) in ways that more ahistorical leaning disciplines have failed to uncover.

Dominant methodologies: the prioritization school is most certainly motivated by largely quantitative data, reinforced by extensive field research, that led scientists to alert the world about the ecological crisis at hand (Kütting 2000). It places careful attention on what scientific evidence indicates is required to avoid, or reverse, the effects of humans in general, and commodification of nature in particular, on

terrestrial, marine and atmospheric ecological systems. This task requires understanding, and disentangling scientific evidence about the shorter "shelf life" of land-use caused climate change emissions that also threaten biodiversity, from longer term emissions caused by carbon dependent industrialization processes. Members of this school are also keenly aware of the need to avoid conflating data collected by Type 1 commons scholars on "extent of forest cover" or "tree density" from data on the decline of ecosystem structure and function and species extinction risks.

However, what is important for this book is that while prioritization scholars are motivated by ameliorating problems that became known through ecological sciences applying largely quantitative methods, they also recognize that, for instrumental reasons, effective policy and institutional solutions require sophisticated training in qualitative methods. This is owing to the recognition Type 4 environmental problems – such as the massive species extinctions and climate crisis – are overwhelmed in political and policy processes that emphasize Types 3, 2 and 1 short-term or material interests (Lockwood et al. 2017; Levin et al. 2012). Hence, the prioritization school must always embrace methods best suited to identifying "critical juncture" policy designs capable of ameliorating, or avoiding, the problem in question (Jordan and Moore 2020).This is because Type 4 environmental problems – such as the massive species extinctions and climate crisis – are overwhelmed in political and policy processes that emphasize Types 3, 2 and 1 short-term or material interests (Lockwood et al. 2017; Levin et al. 2012). An emphasis on critical junctures has led prioritization scholars to draw on path dependency analysis as an innovative way to think about the triggers of transformative change pathways (Peters and Tarpey 2019; Hacker and Pierson 2014). In contrast to the economic optimization school that views path dependency as inefficient or utility undermining (North 1990; David 1985), prioritization scholars embrace path dependency processes to think about how to change policy outcomes, values, and feasibility calculations (Lockwood et al. 2017; Pierson 1993; Hacker 2001). Specific techniques include qualitative "process tracing" and "applied forward reasoning" (Bernstein 2000) with which to project forward multiple unfolding causal steps resulting from a particular critical juncture.

Political science research has identified at least two distinct ways in which critical junctures might be triggered: societal and stakeholder conflict[1] on the one hand, and innovative "policy mixes" on the other hand (Howlett 2019b; Peters et al. 2018). The former process cautions political scientists who advocate for stakeholder and intergovernmental processes to be mindful not to inadvertently make decisions that undermines productive societal conflict through the creation of "decoy" deliberative bodies that undermine contestation required for more significant transformations (Dimitrov 2019). The latter turns analytic attention not only to the goals of policy, but the way in which specific calibrations that dictate resources and approaches to compliance, or the specific policy settings that determine behavioural changes, matter not only for identifying micro level "easy to pull", but "hard to reverse" policy levers capable of producing, over a series of steps, desired results.

Given the impossibility of developing generalizable theories about the myriad of possible policy triggers that might be uncovered, this school champions the development of diagnostic design questions with which to guide inductive deliberative exercises for generating policy innovations (Levin et al. 2012; Young 2019). For these reasons prioritization scholars (Hildén, Jordan, and Huitema 2017) caution against the dominance of "experimentalist" policy designs favoured by many self-labelled "data-driven" "empirical social scientists" (Cole 2015) because evidence of transformative effects appears far too late to inform policy design. They are particularly concerned that large-N experimentalist approaches, and learning that results, tend to implicitly bias Types 1, 2 and 3 because empirical evidence for these problems is much more likely to appear within weeks or months following the intervention at hand. It is, arguably for these reasons that climate focused learning experiments tend to focus on human adaptations to, rather than lessons for mitigating, Type 4 environmental changes (e.g.Armitage et al. 2009).

For these reasons, the prioritization school requires significant expertise and training in conceptual and

reflexive analysis, as well as comparative policy studies, with which to foster inductively generated policy designs (Howlett and Cashore 2009; Howlett 2019a). Owing to concerns of biasing a quest for (Type 2 and 3) universalist theories, quantitative empirical analyses play an important but secondary role in the search for policy designs capable of identifying change levers. It is for these reasons ironic, if not tragic, that GTC and subsequent meta reviews on theories of the policy process championed by Sabatier (Sabatier 1999) and (Ostrom 1999) ignored or dismissed historical institutionalist scholarship precisely because it did not seek generalizable frameworks (Sabatier 1999, 5).

Theory building project: this school turns to two distinct ways in which critical junctures might be triggered: societal and stakeholder conflict[6] on the one hand, and innovative "policy mixes" on the other hand (Howlett 2019a). The latter turns analytic attention not only to the goals of policy, but the way in which specific calibrations that dictate resources and approaches to compliance, or the specific policy settings that determine behavioural changes, can identify "easy to pull" micro level levers capable of producing, over a series of steps, desired results. While these methods are certainly advanced in some strands of policy sciences, they are, as I argue in the introduction to this book, the required analytical capacities to achieve these outcomes are poorly tended to, and with few resources deployed, compared to the enormous set of resources deployed to "double down" and advance Type 3, 2 and 1 reinforcing methods. It is for these reasons that the last half of this book carefully walks us through how to build analytical capacities for integrating the complete range of causal knowledge for advancing Type 4 policy pathways.

Effectiveness: for all these reasons the Common's school is anchored in achieving desired policy objectives facing a clearly specified, and measurable, "on the ground" challenge whether it is about maintaining the long-term supply of utility enhancing fish for a poverty stricken community, timber management in 1000s of forest dependent communities across India, or long term supplies of clean water facing local urban areas, there is no success, and no sense of accomplishment, if the specified issue at hand becomes degraded or collapsed. This is important because its attention on revisiting theories and prescriptions if they don't cause the resource good to be maintained (rather than simply managing degradation more slowly) is not only a critically important aspect of this School for Sustainability scholarship policy and governance, its approach to "on the ground" effectiveness in managing the particular Type 1 problem at hand, can, and must, be emulated by Type 4 problems that consumer the interest, and attention, of the Prioritization school.

Birds and moles: the prioritization school recognizes that its sequentialist morality can create "collateral damage" moles. This has led to a critique that ranking environmental problems as highest on the pecking order is "elitist" (Farrell 2020) and even undermines the sustainable development of some of the world's most poor and vulnerable populations, including indigenous communities and rural livelihoods. However, prioritization school scholars respond it is often possible to grant lexical ordering to the environment that can, indeed, subsequently ameliorate social and cultural challenges (Clapp and Dauvergne 2005), just not in the way envisioned by economic optimization or compromise schools that, in so many cases, result in "lose/lose" undermining outcomes for ecosystems and local cultural traditions (McAfee 1999). While the prioritization school within sustainability studies has thought carefully about how to handle external moles, it has largely failed to explicitly assess the problem of internal moles: i.e., how to conceive of the range of Type 4 problems with which to grant lexical status. Intriguingly, GTC called for such thinking, arguing that while biological and ecological systems were different from those covered in GTC, they, too, could be enhanced by similar inductive attention to the features of the problem at hand species (Ostrom 1990, 25-26).

Attention to these matters leads to the identification of four different ways in which the prioritization school thinks about what environmental challenges to prioritize (Table 3). The *holistic* approach is motivated by scientific evidence that, in some cases, the structure and function of an entire ecosystem is

required to maintain the viability of an individual species. The end point for lexical ordering means that both the species and eco-system must be granted first order status if each of them is to be effectively addressed. In these cases, data are collected to assess what amount of extractive activity, including eco-friendly logging practices, might be permitted without compromising either individual species or the broader ecological systems (Lindenmayer and Franklin 2003). Intriguingly, GTC appeared to favour this approach when advocating for the identification of relevant proxy species (Ostrom 1990, 25-26) with which to guide inductive policy and institutional solutions for advancing ecosystem sustainability.
In contrast, the *targeted* approach draws on scientific evidence – especially but not exclusively from tropical and subtropical climates – that some species, including non-charismatic invertebrates, live on such small parcels of land that any act of extraction will almost certainly result in their extinction (Grove 2002; Wijedasa et al. 2017; Nair 2007).

The *comprehensive* approach targets maintaining ecosystem structure and function in the face of some degree of "inevitable" species loss given human disruptive practices. The overarching concern is to identify the "limits" or "carrying capacity" of natural systems. This approach is behind the Paris Accord's 1.5°C goal – which, drawing in the IPCC's scientific review, explicitly accepts some degree of ecological degradation caused by greenhouse gases, but at a level that is expected to avoid cause catastrophic changes in ecosystem structure and function. The *technocratic* approach is motivated by the scientific evidence concerning species decline given inevitable ecosystem degradation. This approach has resulted in policy innovations from the creation of seed banks to store genetic material from degraded or destroyed ecosystems, the strategic placing of feeding sites for migratory birds, and even the creation of zoos and wildlife conservation areas. It also captures the motivations of those political scientists who, working with engineers, focus efforts on developing innovative technologies to reduce carbon footprints while explicitly treating the 1.5/2°C goal as infeasible.

Table X.X:
Four Approaches to the Tragedy of Type 4 Species Extinctions

		Ecosystem wide	
		Yes	No
Absolutist	Yes	**HOLISTIC** • Species dispersed across ecological systems e.g., managing vertebrate species such as wolfs and owls viable in boreal and temperate forests	**TARGETED** • Species within narrow habitats e.g., avoiding extinctions of invertebrates such as spiders in tropical forests
	No	**COMPREHENSIVE** • The science of ecological systems in midst of species decline. e.g., management within Borneo's Peatlands, US grasslands	**TECHNOCRATIC** • The science of saving some species in midst of ecosystem decline. e.g., seed banks, bird feeding stations, planting trees in non-tree ecosystems, zoos

Table created by Cashore and first published in Cashore and Bernstein (2022)

These distinctions help qualify contradictory findings and arguments in sustainability studies and science. First, there are *no synergies* for the targeted approach because it is concerned with those empirical cases in which any human satisfying activity will result in extinction. This means that anyone declaring that there are mostly synergies across sustainability objectives is advancing a metaphorical conclusion that must, by definition, ignore the most relevant and important data on species extinctions. For these reasons scholarly efforts that find, or assert, synergies, ignore, or bypass, this empirical evidence – often in the name of "evidence based" or "science based" targets (Khan 2024; Clancy 2020). Second the holistic approach narrows consideration of synergies to those utility enhancing activities that, following scientific evidence, are not likely to threaten the viability of ecosystems and its species. Third, while the technocratic orientation has the most synergies across problem types, its acceptance of ecologically degraded world is hardly the outcome that most Type 4 environmental sustainability scholars would find celebratory.

Commons School (Type 1 reinforcing)

- **Problem contingent?**: **Yes.** commons tragedies
  - careful attention to the fundamental features of a clearly identified and measurable collective action challenge, most notably a specific "tragedy of the commons", that would need to be taken into account to understand its causes and potential solutions
    - e.g. the absence of rules excluding access to grasslands, or the ability of fish to swim making excludability difficult
- **Utility dominates? Yes** utility undermining collective action problems
- **Solution orientation**: the development of a long-term oriented economically optimal governance institution capable of avoiding a clearly specified "on the ground" utility undermining problem, highlighted by resource collapse
- **Ethical motivations:** genuine compassion to help communities avoid catastrophic collapses of resources
- **Dominant disciplines**: economics, rational choice political science, some strands of anthropology
- **Dominant methodologies**: quantitative data for developing design principles, supported by qualitative ethnographic case studies
- **Theory building project**: **contingent**
  - causal and prescriptive insights narrowed to only cover a specific problem, or a specific class of problems
    - e.g. design principles for "common pool resources" (CPR) management
- **Effectiveness:** outcome must avert long term collapse or degradation of the resource in question
- **Moles:** This school thinks deeply about avoiding moles by applying the wrong kind of collective action frame. However the schools silent on what to do when solving a particular collective action tragedy leads Type 2 moles (such as improving timber sustainability when another agricultural crop would have been more lucrative: Type 3 moles (such as undermining legitimacy of political processes, and Type 4 moles, such as species extinctions that occur from utility enhancing practices.

Prioritization school (Type 4 reinforcing)

- **Problem contingent?**: **Yes** careful attention to the fundamental features of a clearly identified and measurable environmental, social, or cultural problem that would need to be taken into account to understand its causes; and potential solutions
  - Specific kinds of social and cultural outcomes such as slavery or the loss of longstanding indigenous practices and cultural traditions;
  - environmental tragedies such as species extinctions or the catastrophic ecological effects of climate change or collapse of Peatland ecosysetms
- **Utility dominates? No .** This school is preoccupied with moles caused by effective Type 1,2 or 3 behaviors and policy solutions. These include:
- **Solution orientation**: ***sequentialism*** or lexical reasoning in which problems are ranked according to their importance such that managing lower ranked problems is limited to those policies and behaviors that to not undermine higher ranked problems
- **Ethical motivations:** averting the worst moles caused by Type 1, 2 &3 motivations, from slavery to species extinctions to ecological catastrophes of a warming world
- **Dominant disciplines**: law (especially environmental and human rights strands) some strands of political science (especially historical institutionalism) political philosophy, philosophy, sociology and ecological economics
- **Dominant methodologies**: path dependency analysis, historical institutionalism, process tracing, anticipatory policy design, applied forward reasoning, some strands of scenario analysis and modeling
- **Theory building project**: **contingent**
  - causal and prescriptive insights narrowed to only cover a specific problem, or a specific class of problems
    - Such as policy pathways for overcoming the tragedy of "super wicked" problems
- **Effectiveness:** achieving higher ranked policy objectives that cannot be undermined by lower ranked problems
- **Birds and moles :** This school thinks deeply about moles caused by Type 1, 3 and 3 frames by arguing for an explicit "lexical" or "sequentialist" approach to problem solving such that lower ranked problems do not cause higher ranked moles. However, the school is largely silent on how to engage in a ranking system, and how to approach lower ranked moles that may undermine legitimacy of the political system and economic opportunities.

Optimization school (Type 2 reinforcing)

- **Problem contingent?**: **No** applies a framework to adjudicate the problem at hand that includes consideration of a range of other problems. Only taken into account if doing so produces policy innovations that maximize aggregate utility of individuals, organizations or states, rather than being drawn on to assess what is required to effectively manage a specific "on the ground" problem per se.
- **Utility dominates? Yes**
  - suboptimal society wide utility undermining outcomes
    - measured through income and wealth, or by converting non-economic outcomes, such as reduction in human satisfaction from losses of nature and cultural traditions into utility equivalents (often through "willingness to pay" surveys).
  - international agreements that fail to maximize state preferences (whether for relative or absolute gains)
- **Solution orientation**:
  - a policy action (or inaction) that is expected to maximize aggregate utility; an innovative global governance arrangement that is expected to maximize state preferences
    - These "***optimal***"solutions will determine not only ***how***, but ***whether***, a specific "on the ground" problem can be effectively managed
- **Ethical motivations:** facilitating individuals, organizations and countries to maximize their long-term self-interests, which they believe will causally produce a plethora of social and environmental goods from income to housing to schools to conservation
- **Dominant disciplines**: economics (classical & behavioral), political science (rational/public choice; behavioral); international relations (neo-realist and neo-liberal); some strands of philosophy (esp. utilitarianism)
- **Dominant methodologies**: cost-benefit analysis including econometrics modeling, game theory
- **Theory building project**: **universal**
  - adherents seek to generate universal (aka "general") theories of self-interested motivations of individuals; organizations or nation states
- **Effectiveness:** maintaining highest level of society wide utility maximization
- **Birds and moles:** This school thinks deeply about moles causes by Type 1, Type 3 and Type 4 framing though its emphasis on "internalizing externalities" in which it anticipates as many moles (and synergies) as possible, works to grant them utility equivalents to avoid utility undermining moles. However the school is silent on what to do about moles that are created by applying Type 2 reasoning, such as loss of biodiversity and cultural traditions.

Compromise school (Type 3 reinforcing)

- **Problem contingent?**: **No** Applies a pre-existing framework that is independent of the problem structure or taking into account what is required to effectively manage the problem at hand qua the problem at hand.
- **Utility dominates? No** this school takes into account a range of interests, organization and values in its goal to achieve consensus or legitimate decisions. It often does so by seeking to reduce conflict among a subset of individuals, organizations or nations states that undermines legitimacy and the principles of "good governance" rather than utility er se
- **Solution orientation**: incorporating different stakeholders, values and interests into governance and policy making that are deemed to be "fair", "equitable" and "balanced", and that is expected to produce "legitimate" compromises across economic, social, environmental, and cultural values
  - the resulting policy decision will determine not only ***how***, but ***whether***, a specific "on the ground" problem can be effectively managed
- **Ethical motivations:** the building of legitimate and appropriate governance arrangements that unite local, subnational, national and transnational communities
- **Dominant disciplines**: political science (comparative public policy, some strands of political philosophy), sociology, some strands of geography, anthropology, philosophy and law
- **Dominant methodologies**: multistakeholder dispute resolution, historical and qualitative analysis, reinforced by quantitative methods for testing general propositions
- **Theory building project**: **universal**
  - adherents seek to generate universal (aka "general") theories about how to go about building or maintaining legitimacy, and or design dispute resolution procedures and multistakeholder policy dialogues capable of producing agreement among competing interests and values
- **Effectiveness:** achieving long-term agreements for action that takes into account a range of factors beyond the specific problem or even problems at hand
- **Birds and moles :** This school thinks deeply about moles caused by Type 1 and Type 2 frames by an effort to balance different interests and motivations in ways that create political legitimacy and good governance. It however, it does not offer a theory about what to do when legitimate processes produce decisions that undermine Type 1 or 4 problems

In sum, these distinctions lead to the recognition that [illegible] but competing, ethical foundations (Cashore and Bernstein 2022) that underpin each school's transformative policy design project (Visseren-Hamakers et al. 2023). These ethics, in turn, carry profound implications for how to think about, and address, inevitable synergies and "whack-a-mole"[1] effects (Levin et al. 2012), that can result from a highly effective policy design. Notably, the Compromise, Optimization and Commons schools narrow the ability to address problems conceived of through a Type 4 lens to those that are synergistic with Types 3, 2 and 1 conceptions (Cashore and Bernstein 2022). The framework has been applied to show how these (often) unconscious biases, and the metaphors they produce, can result in policies that are inconsistent with the scientific evidence about

behavioural changes necessary for effectively managing critical problems including the climate and biodiversity crises.

Coders may find it useful to note that Cashore argues that these biases explain the eyebrow raising conclusion of leading sustainability governance scholars that, despite widespread evidence that effective policies governing sustainability goals championed by Commons, Optimization and Compromise schools ***caused*** climate and species extinctions moles (Nilsson and Weitz 2019; Independent Group of Scientists 2019; Khamila et al. 2022), synergies among environmental, social and economic goals "far exceeds the number of trade-offs" (IPCC 2018, sec. 2.5.3) and that, where whack-a-mole effects do occur, they can be converted into birds (synergies) through careful design of transition portfolios (IPCC 2018, 33). In contrast, the Prioritization school's internal approach to adjudicating moles does not leave it to chance that synergies can be found with other sustainability goals and objectives. Instead, it fosters an analytical technique that begins, and ends, with lexical ordering: i.e., its starting point is that second order moles that result from successfully managing higher ranked first order problems must be treated ***sequentially***, and only in ways that do not undermine higher ranked problems. Uncovering these differences is profoundly important because they make clear that only the Prioritization school's adjudicating framework is "fit for purpose" for identifying effective and unconditional solutions for ameliorating the ecological effects of climate change, species extinctions, and, for that matter, the loss of indigenous peoples' traditional livelihoods.

The benefits of unpacking the key dimensions of these schools is that that they uncover what is often a hidden causal relationship: i.e. that while deeply held problem frames and causal beliefs are traced back to each school's distinct moral philosophies, these origins have been hidden by never ending debates about which school's methods produce the most "rigorous" research results. Cashore argues that the tragedy in this masquerading of moral beliefs as objective methods has occurred so slowly and subtly over time, and diffused across scholarship, practice and policy design, that it has largely gone uncovered, let alone systematically explored. In other words, debates about methods – reinforced by "evidence-based" and "data-driven" metaphors - have become so prominent, and each school defined by them, that we have largely lost sight of the competing ethical obligations that produced the methodological biases in the first place. He argues that failure to be explicit about these different approaches to, and effects of, their whack-a-mole outcomes, helps explain the disconnect between applied scholarly and practitioner efforts to address Type 4 environmental problems in general, and SDGs 13 (climate) and 15 (biodiversity) in particular.

B. CODING CHALLENGES

The above review will most certainly facilitate, and place in context, efforts to code qualitative documents against the 4PTs. This facilitates, rather than overcomes, several challenges that a code book must incorporate. First, while some documents may seem straightforward, others may provide a nuanced or even contradictory account that makes sorting more challenging. The solution to this is to provide, and justify, systematic protocols to facilitate sorting of what we refer to as "ambiguous" cases below. Second, different types of documents will need slightly different ways to disentangle the 4PTs. For example, our careful review of journal articles on sustainability policy and governance requires laborious coding of highly complex articles. Our original focused on the Journal of Regulation and Governance was used to both revise our 4PT distinctions, and then to go back, recode, and assess intercoder reliability. However, our approach to coding abstracts for course syllabi at Yale's environment schools was much swifter and faster – as we only needed to see whether the course championed a utility outcome or not, and whether it was focused on applying its approach to specific on the ground problems. Both coding exercises permitted the development of dominant "key words" within each both deductively and inductively. This

means that one solution for accelerating coding through quantitively analysis of text is to identify the prevalence of different key words associated with one problem type over the others. Of course this approach is not perfect because sometimes some concepts such as "cost effectiveness" analysis can be used to achieve any kind of Type 4, 3 2 and 1 outcomes, while "cost benefit" reinforces Type 2. It may be difficult for using a key word approach alone to therefore uncover these nuances. Hence, while this approach may not be as nuanced as the human coding orientation, it promises to provide more swift data that itself, will be useful. And as along as standardizes approach is taken, it will still produce useful data even if that set might be slightly different than if laborious human coders had applied our more nuanced approach. Finally, and to integrate these different efforts, our ultimate goal is to draw on AI techniques that might best replicate human coding, but in much less time.

Another challenges concerns whether an article is simply categorized by asking and answer the two distinguishing questions, or, to save, time, whether a sorting approach can be undertaken through a "branching tree" metaphore: i.e. first ask: "utility" if so, (1 or 2) then ask "problem contingent?, if so 1 or 4. Under this approach Type 3 emerges because other types are excluded, rather than affirmatively coding for Type 3. The problem with this approach – and which tripped us up initially – was that Type 3 was more likely to include ambiguous cases and or cases beyond our universe (i.e. they did not actually pertain to policy and governance). As a result, we had to regroup and instead encourage coders to identify each one according to the two categories, and then to note cases where they simply could not figure out how to best apply the code book (which meant we needed to revise and provide better guidance) and/or be clearer about whether a paper was in or out of our universe. For example, we excluded purely science based classes from our review of Yale's environment school, since most science based classes provided basic knowledge about problems – which might appear to be Type 4- but because they didn't identify policy or governance solutions – where actually not part of our universe of analyses.

C. Coding Application

Here was our original approach. One the coder has answered, and justified, the five key questions in Section D, they must now input the data into the master spreadsheet in the following setps.
.

1) **Is the article in the universe we are exploring?**
    a. **If no, discard and code as such**

NOTE: the next two questions are opposite sides of the same coin. We are choosing to ask the question in two different ways, rather than having a "default" coding approach that could mask ambiguity or uncertainty.

2) **Is the article problem contingent?**
    a. If yes, code as such in the spreadsheet
3) **Is the article championing generalizable theories and conclusions of behaviour of individuals, organizations and states that are meant to apply beyond the problem or class of problems at hand?**
    a. If yes, code as such

NOTE: the next two questions are opposite sides of the same coin. We are choosing to ask the question in two different ways, rather than having a "default" coding approach that could mask ambiguity or uncertainty.

4) **Is the article championing or assuming self-interest/utility/satisfaction of individuals, organizations or states?**

a. If yes code as such

5) **Is the article offering an approach that assume multiple motivations of individuals, organizations or states that are not reducible to self-interest/utility/satisfaction conceptualizations?**
   a. If yes, code as such

Decision tree

We will then be able to create an automated spreadsheet that produces the code based on these answer. Any contradictory answers will be labelled automatically as ambiguous.

The coder will then need to reassess answers to the questions to reflect on whether a coding error was made. If no coding error was made, then the coder will enter a "confirm" ambiguous classification (NOTE: we expect few of these now with the clear code book. We will discard ambiguous papers for purposes of our analysis, unless they become "unambiguous" through a revised code book discussion)

D. Application beyond scholarly text analysis

This section will review challenges and our solutions to non-scholarly articles.

a) World bank evaluation reports

b) Scholarly syllabi

c) News stories

d) Press releases/ Policy announcements

e) Mission statements: NGOs, firms, government agencies

E. Coding through data scraping and Machine learning

While key word approaches will never be as nuanced as manual coding, they do allow for application of greater data driven quantitative techniques and outcomes that should also prove useful for basic and applied research. We understand that quantitative classifications will be less nuanced than the qualitative assessments. In order to move them somewhat closer to a qualitative review, the section on key words below for advancing the quantitative "machine coding", which draw on the, and be justified by, key words that emerge from the qualitative review.

F. **Key words approaches**

5. **inductive**

6. **quantitative analysis of key words based on coded articles**

G. Machine learning/AI

To be filled int

H. Appendix A: Coding ambiguous articles from the Journal of Regulation and Governance

Article 1: Assessing the institutionalization of private sustainability governance in a changing coffee sector

Authors: Janina Grabs

Abstract: The potential of transnational private governance initiatives to constitute effective alternatives to state-led regulation of global value chains rests on their ability to scale up and become institutionalized in a given sector. This study examines whether such institutionalization has occurred in the coffee sector, the commodity with the most widespread adoption of certified products and over 30 years' experience of private governance, and tests hypotheses on facilitating and inhibiting conditions. It finds that while norm generation around responsible supply chain management and the organizational institutionalization of standard-setting bodies is well advanced, the practice of internalizing social and environmental externalities through the routinized production and purchase of higher priced certified goods continues to be questioned by industry actors. Indeed, conditions that favored normative and organizational institutionalization, such as high levels of industry concentration, product differentiation, and deliberative interaction, are shown to represent barriers to the practice-oriented institutionalization of market-driven regulatory governance.

Comments from previous researchers: Type 2 or Type 3.

The paper examines how private governance for sustainability can be institutionalized using the coffee sector as an example.

- It explores three ways/forms of institutionalizing: norms, organizations, and practices.
- The paper identifies key reasons for the lack of institutionalizing of private sustainability governance in the coffee sector.
- The paper does not discuss how and whether private governance can address environmental problems or address the structural features of the problem.
- It is difficult to ascertain whether the paper uses an economic rationale, which could classify it as Type 2 or a non-economic rationale, which could classify it as Type 3.
- Key reasons for lack of institutionalizing PSG in the coffee sector include the non-existence of a stable political community of practice, the non-existence of sustained demand for certified products, highly politicized issue areas, existence of hegemonic power structures, and passive revolutions.
- Large MNCs/retailers find certification to be an expensive tool and have created their own certifications/labels instead of depending on third-party certifiers.

Problem classification: Type 3. The problem of how private governance for sustainability can be institutionalized focuses on balancing the interests of various stakeholders, rather than addressing the environmental problem itself.

Article 2: **Authoritarian but responsive: Local regulation of industrial energy efficiency in Jiangsu, China**

Authors: Junming Zhu, Marian R. Chertow

Abstract: Regulatory behavior and effectiveness in authoritarian settings are subject to alternative characterizations. By tracing enforcement processes through a variety of case studies, this article proposes and refines a new model, at least with respect to energy efficiency regulations in China: authoritarian but responsive. Local rulemaking and operationalization is authoritarian, with strong and coordinative bodies of regulation, strategic plans, and active involvement of local authorities. Local authorities, however, often find themselves facing a welter of laws imposed on companies that create competing priorities for

these local officials who then must struggle to find pragmatic solutions. On numerous occasions, such satisficing behavior by local officials makes them responsive to the performance and demands of regulated firms. Embedded in the decentralized authoritarian context, the authoritarian but responsive approach is found to be a rational choice of local governments and different from previous conceptualizations. It helps local governments coordinate across a diverse array of regulatory issues. Drawing on environmental enforcement in China, the model provides consistent explanations for the seemingly changing and discretionary enforcement incidents.

Comments from previous researchers: Type 3 or Type 4.

- The paper explores governance within an authoritarian system, where local regulators compete with different interests and a strong central regulator that demands action on pollution and energy demands.
- The environmental policies in China are given type 4 status due to the strength of the central government.
- The paper draws on a type 3 analysis to understand how local regulators maneuver within this framework.
- The paper suggests that energy regulators should adopt a combination of Type 4 and Type 3 approaches, with a focus on improving China's energy efficiency.
- Type 4 policies include closure of old plants with low energy efficiency, electricity quota, mandatory upgrades of tech, etc.
- Type 3 approaches suggest responsive enforcement to improve overall regulatory effect, with direct preferential enforcement of less stringency compared to other firms and alleviation of previous punitive enforcement.
- Local authorities were found to be flexible and responsive to industrial performance when enforcing mandatory policies, leading to better compliance and improvement in energy efficiency.
- Regulators were more flexible based on the firm's energy efficiency and other local contributions such as treatment of local waste, employment levels, tax revenue, etc.

Problem classification: Type 1. The problem focusses on the impacts of governance regimes on specific environmental issues, in this case "industrial energy efficiency".

Article 3: Transparency in transnational governance: The determinants of information disclosure of voluntary sustainability programs

Authors: Philip Schleifer, Matteo Fiorini, Graeme Auld

Abstract: The rise of "new" transnational governance has intensified debates about a lack of accountability in global politics. Reviewing the mechanisms through which transparency can foster accountability beyond the state, this article explores the determinants of information disclosure in the field of transnational sustainability governance. Examining the institutional design of 113 voluntary sustainability programs, we find a positive correlation between the involvement of public actors and information disclosure. In contrast, the role of civil society is more ambiguous. There is no statistical support for arguments linking nongovernmental organization participation to increased transparency. At the same time, our analysis reveals a robust correlation between civil society-led metagovernance and information disclosure. Moreover, we find that crowding has a negative effect on transparency, whereas normative peer pressures have no influence. At a broader level, the analysis reveals a lack of "deep transparency" among transnational sustainability governors. This limits the scope for transparency-induced accountability in this policy domain.

Comments from previous researchers: Type 2 or Type 3.

- The article uses a quantitative research approach to analyze 113 voluntary sustainability programs and their level of transparency.
- The focus is on accountability among actors and information disclosure.
- The analysis is Type 3 and does not address structural features of the problem.
- The article explores the possible causes and determinants of transparency and information disclosure.
- The impact of VSPs and higher information disclosure on environmental outcomes is not explored.
- The theory of club goods is used to analyze information disclosure patterns of VSPs.
- The article could be classified as Type 2, but Type 3 seems like a better fit.

Problem classification: Type 3. The problem focuses on accountability among actors in transnational governance.

Article 4: Symmetric and asymmetric motivations for compliance and violation: A crisp set qualitative comparative analysis of Chinese farmers

Article 5: Judicial enforcement deputies: Causes and effects of Chinese judges enforcing environmental administrative decisionss

Article 6: Fragmented or cohesive transnational private regulation of sustainability standards? A comparative study

Authors: Luc Fransen, Thomas Conzelmann

Abstract: Literature on private regulation recognizes the proliferation of competing regulatory organizations and approaches in various industries. Studies analyzing why fragmentation arises so far focus on single-case studies, the exploration of single variables, or variation in types of fragmentation. This article analyzes why in certain industries and for certain issues regulatory organizations proliferate, while in others a single regulatory organization emerges which covers the entire industry. Through a comparative case study of private regulation of sustainability standards in the forestry, clothing, IT-electronics, and chemicals industries, we show how a combination of low industrial concentration, civil society involvement in governance, and stringent standards of a first-moving regulator offer the strongest explanation for a fragmented private regulatory field, while high industrial concentration, business-driven governance, and lenient standards of a first-moving regulator lead to cohesive regulation.

Article 7: Dynamic governance interactions: Evolutionary effects of state responses to non-state certification programs

Article 8: Transnational governance and the re-centered state: Sustainability or legality?

Authors: Tim Bartley

Abstract: After two decades in which transnational governance of production processes has typically meant voluntary subscription to privately developed standards, some transnational rulemaking projects are promoting mandatory compliance with law. The emerging timber legality regime is one example of this, and scholars' efforts to theorize this regime have produced provocative new analyses of interactions between public and private standards. Recent analyses, including those in this issue, predict that the new legality regime will bolster voluntary initiatives that certify sustainable forests. Based on research in Indonesia and China, I argue that this prediction is questionable and that the rise of the timber legality regime could constrict, rather than expand the space for global private authority. Further, I argue that it would not be entirely a bad thing if the legality regime overtook sustainability certification. Behind these specific arguments are general perspectives on how domestic circumstances shape transnational business

governance and on the role of states in pluralistic fields of governance – both issues that are obscured by more architectural approaches.

Article 9: Orchestrating sustainability: The case of European Union biofuel governance

Authors: Philip Schleifer

Abstract: This article provides an empirical analysis of orchestration – that is, the initiation, support, and embracement of private governance arrangements through public regulators – in the field of European Union biofuel governance. It examines the emerging sustainability regime and shows that orchestration has been extensively practiced. Regulators in the European Union have used a range of directive and facilitative measures to initiate and support private biofuel certification schemes and to incorporate them in their regulatory frameworks. This has given rise to a hybrid regime in which public and private approaches are closely intertwined. Discussing the benefits and complications of engaging with private biofuel sustainability governance, the article's findings point to a partial failure of orchestration in this policy area.

Comments from other researchers: Deciding between Type 2, Type 3 and Type 4

- Type 2 argument was based on it being more focused on the economic rather than social – and the question lies in whether the economic and the social side was more dominant?
- Type 3 mentions that it was not only a concern for the environment but also a social safeguard perspective. However, it was not specific. Good governance as a compromise school approach and the legitimacy of the government mechanisms - 'reducing costs and increasing reach'.
- Type 4: critique of sustainability itself? **Not type 4: must clearly show an alignment of the problem and thinking about the solution.**
- It is a Type 2 problem with a Type 3 solution – and we want to distinguish the framework of analysis from the tool used to achieve it.

It is a Type 3 paper – 'as a result, orchestration appears to have undercut rather than promoted good private governance in the biofuel industry.' *orchestration* tends to be type 2 – game theorist approach - rather than type 3).

Article 10: Playing games of governance: How and why Fair Trade pioneers evade corporate capture

Authors: Anna Hutchens

Abstract: The concept of power in political governance has traditionally focused on domination and the preservation of the status quo. In an economic context, institutional and organizational studies have expressed growing interest in the dynamics of agency and institutional change, captured in the concept of "institutional entrepreneurship." In the context of global free trade, the Fair Trade movement's experience shows that ongoing institutional entrepreneurship is important for entrepreneurs to transcend absorption by corporate hegemony. In this article I examine the capacity for agency in market institutions through the lens of "defiance" to illuminate the imaginative "game players" who evade institutional capture in the evolution of market governance.

Comments from other researchers: Deciding between Type 2 and Type 3
Type 2 because it mentions but does not specifically look into the social mechanisms. It lists a plan of action unlike Article 9 which has at least considered social mechanisms at hand. In the conclusion - defiance offers us a lens of capturing and understanding game playing in political governance and, in turn, the possibility of institutional change in global markets. For fair trade movement to survive, it has to be consolidated with growing amounts of entrepreneurship - throws me off a bit that its type 3 but it is not specific throughout the paper.

It is not type 2 because from the tool we cannot tell the type of problem and the means to achieve a problem. Type 3 – talks about social mechanisms with an agency to defy the current institutions. Fair trade considers the community, social welfare in market expansion. It is type 3.

Article 11: Knox-Hayes, "Negotiating climate legislation: Policy path dependence and coalition stabilization" (DRAFT)

Ben's notes:
First it does not always follow that an application or assessment of game theory or even economic valuation will mean the paper is Type 2. This is because the paper may be assessing whether the theory works to explain something, rather than assuming the theory is correct. Second, just like cost-effectiveness analysis improved the compliance and reduce costs of Type 4 acid rain regulations through the development of a "cap and trade" system, an application of game theory or ACF/sabatier theory could be used to assess how coalitions might influence, or not, "on the ground" outcomes such as 1.5/2 degrees. Similarly, knowledge about how dialogues work a la compromise school could be reshaped to think about type 4 problems (which is how many have applied Sabatier/ACF)

- So game theory and costs and benefits are clues could be type 2 but not always (as the code book discusses)
- Likewise assessing coalition formation to think about effects of problem solving would be expected to draw on all kinds of theory of support, including game theory – but that wouldn't be its ultimate interest and it owuldn't be realying on game theory to tell us WHETHER the problem could be solved – only instead to understand coalition formation
- Second,
- the article is very clear that its results are not consistent with ahistorical motivations assumed by RC theory

- *"Policy path dependence suggests that theory alone is insufficient to predict policy outcomes; policy results depend strongly on prior policy efforts, historically and socially contingent coalitions, and the resulting framing of policy possibilities.*
- *At the same time, and ironically, the article says it can model these interactions*
- *"*If the conditions of prior policy and coalition formation are adequately understood, the modeling of coalition formation suggested here could be used to predict or project future policy trajectories. The model could also be used to understand current policy dilemmas and to more quickly find the equilibrium or range of equilibriums that will stabilize the field of conflict.*"*
- *The ironies and contradictions of these two quotes can be discussed later* 😉
- Third, result we need to distinguish the instrumentalist orientation of the paper ("how can we understand coalition formation to improve climate outcomes?") from the findings itself.

  For example even though the article found that ahistorical RC couldn't explain outcomes, what if it did find this? Would that still be biased, or would it instead tell of the "impossibility" of finding non RC behaviors? My feeling is that because we know the world isn't explained solely by RC behaviors, that this "finding" is an example of bias, rather than an empirical observation (granted same could be said for historical critical juncture work, but at least these folks are open to finding broader sets of motivations that can and do exist).
- For all these reasons this article falls outside Type 2 thinking, even though influenced by it. But, does it go enough in the direction of Type 4? Or does it provide answers that would limit an ability to solve the climate crisis qua the climate crisis?
- I again go back to this conclusion: "*This could have a considerable benefit inaccelerating the rate of policy development under conditions requiring quick action, such as climate change."*

"Quick action" is clearly the first key feature of climate change. At the same time the problem is weakly tended to and rather abstract.

So I can see coding on both Type 4 and Type 3. At the same time, because the article discusses the clean air act, and discussions pollution as its focus, and because it does shed light on how to solve, rather than simply seek consensus or compromise about climate action, at the end of the day, it seems **best classified as type 4**

Article 12: Regulation and voluntarism: A case study of governance in the making

Authors: Tamar Barkay

Abstract: In this article I analyze a multi-stakeholder process of environmental regulation. By grounding the article in the literature on regulatory capitalism and governance, I follow the career of a specific legislative process: the enactment of Israel's Deposit Law on Beverage Containers, which aims to delegate the responsibility for recycling to industry. I show that one crucial result of this process was the creation of a non-profit entity licensed to act as a compliance mechanism. This new entity enabled industry to distance itself from the responsibility of recycling, and thereby frustrated the original objective of the legislation, which was to implement the principle of "extended producer responsibility." Furthermore, this entity, owned by commercial companies and yet acting as an environmentally friendly organization, allowed industry to promote an anti-regulatory agenda via a "civic voice." The study moves methodologically from considering governance as an institutional structure to analyzing the process of "governancing," through which authoritative capacities and legal responsibilities are distributed among state and non-state actors. Two key findings are that this process and its outcome (i) are premised on an ideology of civic voluntarism, which ultimately delegates environmental responsibilities to citizens; and (ii) facilitate an anti-regulatory climate that serves commercial interests.

Comments:
Recycyling extended producer responsibility

Two key findings are that this process and its outcome (i) are premised on an ideology of civic voluntarism, which ultimately delegates environmental responsibilities to citizens; and (ii) facilitate an anti-regulatory climate that serves commercial interests.

the emergence of public–private regulatory regimes.

"regulatory capitalism" as the overall matrix within which
various forms of "new governance" come into being (Levi-Faur 2005; Braithwaite 2008).

So far above it is Type 3

But this sentence implies possible Type 4: "Reduce litter and divert recyclable items from the waste stream" although the only issue here is that it is still abstract – what is the problem exactly in "reducing litter" very vague…also "waste stream" also vague…in the direction of Type 4 but not exactly measurable yet…

Recycling corporation undermined original purpose of law (this is same in forest practices boards)- to put costs onto industry

Delegated to individual citizens regime. This technique, however virtuous and in line with the aspirations of new governance arrangements it may be, also raises grave questions concerning the ability of organized commercial interests to use it to shape regulatory frameworks in ways that may undermine business's own responsibility for the public good.

My observations: this article clearly is focused on understanding if a public/private innovation might yield compliance for a specified outcome: recycling etc. The answer is no: that it went in the other direction and industry used its power to fend of requirements to individuals that led to outcomes that were not intended

So this seems to be Type 4 BUT the issue here is that recycling is also abstract: what problem exactly was trying to be solved? It is too abstract to be measurable in terms of an "on the ground" problem. BUT at the same time they are thinking of "on the ground" problems however badly defined, and they are assessing where private or non-state regulatory capitalism can solve problems that they were created to address.

Problem classification: Type 4. So for these reasons I would still give a 4 (not because they found a negative answer but because they were assessing whether these innovations improved compliance and uptake)

Article 13: When risk-based regulation aims low: Approaches and challenges

Authors: Julia Black and Robert Baldwin

Abstract: Risk-based regulation is becoming a familiar regulatory strategy in a wide range of areas and countries. Regulatory attention tends to focus, at least initially, on high risks but low-risk regulatees or activities tend to form the bulk of the regulated population. This article asks why regulators need to address low risks and it outlines the potential difficulties that such risks present. It then considers how regulators tend to deal with lower risks in practice. A body of literature and survey-based research is used

to develop a taxonomy of intervention strategies that may be useful in relation to low-risk activities, and, indeed, more widely. In an article to be published in the subsequent issue of this journal, we will then develop a strategic framework for regulators to employ when choosing intervention strategies and we will assesses whether, and how, such a framework could be used by regulatory agencies in a manner that is operable, dynamic, transparent, and justifiable.
Risk-based regulation is a particular strategy or set of strategies that regulators use to target their resources at those sites and activities that present threats to their ability to achieve their objectives (Black 2005a, 2008; Black &Baldwin 2010).

The companion article assesses whether, and how, such a framework could be used by regulatory agencies in a manner that is operable,dynamic, transparent, and justifiable, and argues that coming to gripswith the challenges presented by lowrisks compels us to rethink our conceptions of risk-based regulation more generally.

Theory of risk: issue is for what purpose?

"Such survey responses, however, focus on the levels of satisfaction of involved parties rather than provide an independent view of strategies and operations."

Problem classification: Type 3. Not enough discussion on implications for actual problems. They are also trying to develop a framework for understanding risk that seems to be applied both beyond problems, but also nuancing different problems. It is going in direction of Type 4 potentially but too abstract to be Type 3

Article 14: Dynamic governance interactions: Evolutionary effects of state responses to non-state certification programs

Authors: Lars H. Gulbrandsen

Abstract: Research has recognized that states enable or constrain private governance initiatives, but we still know too little about the interactions between private and public authority in the governance of various social and environmental problems. This article examines how states have responded to the emergence of forest and fisheries certification programs, and how state responses have influenced the subsequent development of these programs. It is argued that historical and structural differences in the management of forest and fisheries have resulted in divergent state responses to certification programs, but that both trajectories of interaction have led to a strengthening of the non-state program. The article draws upon these cases to inductively identify types of interaction between state policies and non-state certification programs, the causal mechanisms that shed light on interaction dynamics, and the conditions under which state involvement is likely to result in either strengthening or weakening of non-state programs.
Coding:
weakening or strengthening of non-state certification programs
dynamic interactions between public and private governance
how to states influence?
Number of fisheries certified

"The evidence suggests, however, that the FAO guidelines in combination with other government policies and market dynamics have had a positive impact on the development and uptake of the program. First, the MSC has actively employed the FAO guidelines in efforts to enhance the credibility of the program and attract new clients for certification."

"As with the FSC, government support for the MSC has contributed to its impressive growth in recent years. In both cases, state responses to the emergence of certification programs have served to legitimize certification as a credible policy tool that companies and consumers can trust."

This is definitely not type 4: there is nothing about actual on the ground problems, no measure as to what an "on the ground" outcome would be, and only looks at support and competition.

Whether it is type 2 or 3: it seems to me as it doesn't directly incorporate utility/satisfactions – indeed it is looking at government/private sector interactions with implicit assumption that government policy may include some rationales beyond Type 2… it does seem to be trying to develop a theory of public private interactions beyond a particular problem, but not in a way that is clearly rationalized

Problem classification: Type 3

Article 15: The limits and variety of risk-based governance: The case of flood management in Germany and England

Authors: Lars H. Gulbrandsen

Abstract: Risk-based governance is argued by many to hold the promise of amore rational and efficient state, by making explicit the limitations of state interventions and focusing finite resources on those targets where probable damage is greatest. This paper challenges the assumption that risk-based governance has the potential for universal and uniform application, by comparing contemporary flood management in Germany and England. On first inspection, flooding appears to be a paradigmatic case of risk colonizing European policy discourses, with the traditional notion of flood defense giving way to flood risk management in the context of climate change, increasingly frequent flood disasters, political and cost pressures on flood protection, and publicly available Europeanwide flood assessments. Drawing on in-depth empirical research, this paper shows how the role, and even the definition, of "risk" is institutionally shaped, and how the respective institutional environments of German and English flood management practices impede and promote risk colonization. In particular, the use and conceptualizations of risk in governance are variously promoted, filtered, or constrained by the administrative procedures, structures, and political expectations embedded within flood management and wider polities of each country. The findings of this research are important for the design and implementation of supranational policies and regulations that endorse risk-based approaches, such as the recent EU Flood Directive, as well as scholarly debate as to how to legitimately define the limits of governance in the face of uncertainty and accountability pressures.

Coding:

Problem: flooding

Risk based governance poorly suited to addressing " namely that risk-based governance ensures a more efficient use of finite resources" (cost effective)

Increase in floods that caused economic damage

The reforms in the two countries adopted the labels "making space for water" (DEFRA 2005) and "room for rivers" (Bundesregierung 2005a). As the concept of room/space suggests, land-use regulation has become a central part of the emerging flood management approaches (LAWA 1995, 2004; DCLG 2006). Other domains beyond flood defenses, such as commercial insurance for private risk management (LAWA 2004; ABI & Government 2008) and flood risk communication (EA 2006; UBA 2006), have also received attention.

Ok so this is a question attempting to answer whether risk does something in the context of good governance, transparency, and shaping responses to particular problems, in this case flood management.

Now, it is finding that risk plays out different in different contexts, but it isn't really distinguishing this based on the key features of flooding in each country, but different institutional settings. This means it isn't building a grant theory of rationalist thought, and it is reinforcing Steinburg's concept of "resonsance groups" but not based on problem structure.

Therefore it is not type 4. Is it type 3 or type 2? It is not advocating for a rationalist theory so it is squarely type 3

1) Problem classification: Type 3

I. APPENDIX B: Datapoints for coding journal articles

The following information is to be captured for each article/solution that is being evaluated by the coder. Please use the excel coding template (insert link when ready) to capture information.

#	Parameter	Type of Input (or example)	Comments/ References
Basic Information about the article			
1	DOI	e.g., doi:10.1111/rego.12215	-
2	Name of journal	e.g., Journal of Regulation and Governance	-
3	Volume number	Number	-
4	Date	DD/MM/YYY	-
5	Title of the paper	Text	-
6	Abstract	Text	
7	Official key words specified in the article	Text	
8	List of all Authors	Text	
9	Corresponding author's name	Text	If corresponding author is not available, capture first author's details -
10	Corresponding author's affiliated organization	Text	
11	Corresponding author's highest degree	Text	
12	Corresponding author's field of study (highest degree)		
13	Corresponding author's graduating university (highest degree)	Text	
14	Corresponding author's geographic location (Highest degree)		
15	Corresponding author's PhD dissertation (Only for Political Science)		

#	Parameter	Type of Input (or example)	Comments/ References
16	Corresponding author's research interests (Only for Political Science)		
17	Corresponding author's 5-10 publication titles (Only for Political Science)		
18	Corresponding author's CV Link		
19	Categorization of corresponding author's educational background	Choose from drop down • Natural Sciences • Engineering • Environmental Studies • Non-Economic Social Sciences and Humanities • Rational Choice Political Science • Historical Political Science • Public Policy • Economics • Law • Management, Administration, & Finance • Transition Studies • Others	
Diagnostic questions			
20	Does the article fit in the universe of sustainability analyses we seek to assess? (Yes / No)	Yes / No	If No, remove the article from the dataset. If Yes, proceed with coding.
21	Research methodology of the Article	Choose from drop down • Qualitative • Quantitative • Mixed Methods	-
22	What problems or set of problems is the article trying to address?	Text	Explain your understanding of the problem being addressed in the article
23	Problem contingency Do the analysis, conclusions, and theories derived from, and directed to, understanding and/or managing a clearly specified "on the ground" problem or class of problems?	Yes / No	If Yes, then the article is either Type 1 or Type 4 If No, then the article is either Type 2 or Type 3 Refer to Section C2.1.1 and C2.1.2 in the codebook for additional guidance
24	Provide arguments that supports your response to Q23	Text	-
25	Provide some key text passages from the article that support your response to Q23	Text	-
26	Problem contingency Are the analysis, conclusions, and theories generated to apply beyond understanding, and/or managing a clearly specified "on the ground"		If Yes, then the article is either Type 2 or Type 3 If No, then the article is either Type 1 or Type 4 Refer to Section C2.1.1 and C2.1.2 in the

#	Parameter	Type of Input (or example)	Comments/ References
	problem or class of problems? (Yes / No)		codebook for additional guidance
27	Provide arguments that supports your response to Q26	Text	-
28	Provide some key text passages from the article that support your response to Q26	Text	-
29	Utility Do the analysis, conclusions, and theories treat individuals, organizations and states as largely self-interested, satisfaction driven entities that seek to maximize some kind "utility" outcome?	Yes / No	If Yes, then the article is either Type 1 or Type 2 If No, then the article is either Type 3 or Type 4 Refer to Section C2.1.3 and C2.1.4 in the codebook for additional guidance
30	Provide arguments that supports your response to Q29	Text	-
31	Provide some key text passages from the article that support your response to Q29	Text	-
32	Utility Do the analysis incorporate theories and conclusions incorporate an assessment of individuals, organizations and/or states that extends beyond self-interested satisfaction seeking motivations?	Yes / No	If Yes, then the article is either Type 3 or Type 4 If No, then the article is either Type 1 or Type 2 Refer to Section C2.1.3 and C2.1.4 in the codebook for additional guidance
33	Provide arguments that supports your response to Q32	Text	-
34	Provide some key text passages from the article that support your response to Q32	Text	-
35	What are some key words used in the article that support your problem type classification?	Text	***Examples of key words for Type 1:*** • Optimality • Free rider • Shirking • Collective action dilemmas • Efficiency • Optimization • Internalizing externalities ***Examples of key words for Type 2*** ***Examples of key words for Type 3*** ***Examples of key words for Type 4***
36	Based on the above, what is the problem type of the article (Type 1, 2, 3, and 4)	Type 1, 2, 3, and 4 auto-populated based on response to Q23, Q26, Q29, Q32	
37	Difficulty in coding the article and identifying the problem type	Choose from drop down • 1 – Very easy	Score of 1 and 2 can be considered an 'explicit'

#	Parameter	Type of Input (or example)	Comments/ References
		• 2 – Easy • 3 – Medium • 4 – Hard • 5 – Very Hard	**presentation of problem type** **Score of 3,4, and 5 can be considered an 'implicit' presentation of problem type**
38	**Primary coder's name**	**Text**	**-**
39	**Secondary coder's name**	**Text**	
40	**Secondary coder's coding**	**Type 1, 2, 3, and 4**	
41	**Secondary coder's reasoning**	**Text**	
42	**Intercoder reliability**	**Match if Q36 = Q40** **Non-match if Q36 != Q40**	

Sources

Endnote

Sources
Endnote

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Li Ka Shing Professor in Public Management
Director, Institute for Environment and Sustainability (IES)
Lee Kuan Yew School of Public Policy, National University of Singapore
469B Bukit Timah Road, #02-01, Level 2, Li Ka Shing Building, Singapore 259771
tel +65 5166 6195, sppbwc@nus.edu.sg
Web site: https://lkyspp.nus.edu.sg/our-people/faculty/benjamin-william-cashore

[2]Cashore is lead author on this code book. It includes and draws contributions from several colleagues and research associates and assistants. These include Ashwath Dasarathy and Beatrice Riingen who played pivotal roles in writing and application from January to July 2024 and contributions from He Liuyang during and subsequently and nuanced feed back from Augustin Boey. Other researchers who contributed to iterative processes from 2000-2025 including Li Bo Chris Skelton, Ashwath Dasarathy, Kendra Wong, Ayesha Kumar, Li Bo, Mae Chow, Wesley Ng, Beatrice Riingen, Shao Huizhen, Mehtab Ahmed Jagil, Ankur Kumar Chaudhary, Radhika Rao

[3] To be sure there is a long and established literature, some of it circular, that argues if we are motivated to address Type 4 problems, then that constitutes our "human satisfycing" utility values and hence, voila, the optimization school can still address Type 4 problems. Simimarly another argument goes that we just need to re-orient self-interest in the long run and we can address Type 4 problems. Likewise others, engaging in what to me appears to be some kind of circular reasoning posits that " Altruism can be reciprocity-driven, where individuals help others because they feel obligated to return a favor or anticipate future benefits." Which means they argue, as my colleague He Liuyang notes, that "Indeed, evolutionary perspective suggests that altruism may have evolved alongside the impulse to condemn and even punish those who act solely in their own self-interests.". My point is, as elaborated in this book, that none of the underlying ontologies of the Commons, Optimization and Compromise schools can take us on a Type 4 journey. Efforts to stay within their own school – usually owing to methodological training are understandable, but they become part of the Type 4 tragedies documented in this book (Grateful to Liuyang for a very important observation).

[4] My thanks to Robert Keohane for his critical feedback on an earlier version of my argument in which he did not "recognize" my review of the optimization school, given its emphasis on cost-benefit analysis. I have now clarified that environmental economics is the most important strand, but that others, such as his own work in international relations that contributed so profoundly to a different theory of Type 2 state preferences (Keohane and Nye 1989, 1972) than earlier "rationalist" accounts (Snidal 1991), produced incredibly important and practical insights for enhancing global cooperation over a number of problems that would arise on a country's desk at any point in time. I also appreciate Keohane's feedback that the four Types are "all in my head" since that is exactly the point of this book. That there is no way of looking at the world without some kind of cognitive frame of the problem at hand. While some frames, including those advanced by Keohane and colleagues, perform very well for some kinds of problems, it is also the case that they are not, as this book shows, "fit for purpose" for a range of Type 4 problems that continue to vex our planet.

[5] Economists refer to economic welfare and aggregate utility undermining pollution as the "social cost" of degradation. I simply cannot, in good conscience, use the term "social" to describe economic activity, given the confusion this may cause

[6] My thanks to Guy Peters for this observation.